AW82001 Valvetrain Model Analysis

AVL EXCITE Timing Drive — Intake Valve Train, Right Bank

Model file: vtRBint01.etd

Model path: D:\AW82001\5005\ref_Tamas\AW82001_5004_20-Loop1-ModelStatus\Status20260608\excite_td\

EXCITE version: 24.10

Analysis date: 2026-06-10

Focus elements: INTr_HLIF* (Hydraulic Lash Adjuster), INTr_SPG* (Valve Spring)

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1. Model Overview

The vtRBint01.etd project models the **intake valve train of the right cylinder bank** for the AW82001 engine. It is one of four ETD models in the project directory (also: left-bank intake vtLBint01.etd, and exhaust banks). The model covers 8 intake valves (one per cylinder/valve pair), each represented by a complete element chain from camshaft to valve seat.

1.1 Simulation Cases

The model is set up as a sweep over engine speeds. Results subdirectories are named vtRBint01.Ref_C10.Pup_XXXXrpm:

Run directory	Engine speed	Status
Ref_C10.Pup_7000rpm	7000 rpm	Completed (56 MB)
Ref_C10.Pup_7100rpm	7100 rpm	Completed (60 MB)
Ref_C10.Pup_7200rpm	7200 rpm	Completed (60 MB)
Ref_C10.Pup_7300rpm	7300 rpm	Completed (40 MB)
Ref_C10.Pup_7400rpm	7400 rpm	Completed (33 MB)
Ref_C10.Pup_7500rpm	7500 rpm	Completed (33 MB)

The full load-case sweep defined in the caseTable spans 1500–7500 rpm. Gas load data is provided in Loads/Gas1_RBank/FgasIn_Fgas_XXXXrpm.dat for 1500–7000 rpm.

1.2 Cam Profile

Reference cam: Int165_Reference.GID (INT 165° duration specification)

Follower radius: 8.5 mm

Maximum valve lift: 10.0 mm

Natural frequency: 1990 1/s

Maximum contact stress: 1086 N/mm²

Resolution: 0.25°/data point. Contains 48 result columns including valve lift, velocity, acceleration, contact stress, spring force, and mass force.

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2. Valve Spring — `INTr_SPG*` (8 instances)

2.1 EXCITE Element Type

The EXCITE **SpringMacro (SPG)** element models a complete valve spring as a flexible multi-body chain of coil (SPPR) and torsion (CTOR) elements. Key capabilities:

- Progressive spring rate via force or stiffness characteristic table

- Coil-to-coil contact modeled independently from spring stiffness
- Pitch diagram defines non-uniform coil spacing
- Coupled to rigid ground body at both ends

2.2 Model Parameters — `INTr_SPG1` (representative; all 8 instances identical)

EXCITE element: SpringMacro, elem24–elem81 (elements 24, 25, 36, 45, 54, 63, 72, 81)

Geometry

Parameter	Model value	Unit
Spring type	Cylindrical	—
Wire shape	Elliptic	—
Elliptic wire height (lift direction)	2.95	mm
Elliptic wire width (lateral)	3.36	mm
Coil diameter reference	Outer diameter	—
Outer coil diameter	23.22	mm
Free length	45.2	mm
Total coils	9.25	—
End coils (valve side)	1.0	—
End coils (head side)	2.0	—
Active coils (calculated)	6.25	—
Elements per coil	8	—
Total spring mass	32.0	g

Material

Parameter	Model value	Unit
Shear modulus G	79 500	N/mm^2
Young's modulus E	210 000	N/mm^2
Poisson's ratio	0.3	—
Mass density	7 850	kg/m^3

Installation & Loading

Parameter	Model value	Unit
Installation definition	Preload	—
Preload (valve closed)	250.0	N
Installed length (derived from force table)	≈ 36.2	mm
Spring damping (relative)	0.02 (2%)	—
Damping time constant	1.4×10■■	s
Absolute damping	0.001	N·s/mm
Block stiffness	300 000	N/mm
Block damping (relative)	0.30 (30%)	—

2.3 Force Characteristic Table

The spring characteristic is defined as a **nonlinear force vs. spring length** table with 13 data points, capturing the progressive stiffening behavior:

Spring length (mm)	Compression from free length (mm)	Spring force (N)

45.2	0.0	~0
43.0	2.2	55.7
41.0	4.2	108.0
37.5	7.7	209.7
36.75	8.45	233.2
36.2	**9.0**	**250.8** ← valve closed (preload)
32.0	13.2	396.0
31.0	14.2	433.3
30.0	15.2	471.7
29.0	16.2	511.1
27.0	18.2	592.0
26.2	**19.0**	**625.0** ← near full valve lift
24.0	21.2	730.0

Effective spring rate (valve-closed to full-lift, 36.2→26.2 mm): ≈ 37.4 N/mm

2.4 Stiffness Characteristic

The model also stores an explicit stiffness table (tangent rate), derived from the force characteristic:

Spring length (mm)	Stiffness (N/mm)
45.2 (free)	23.9
43.0	23.9
41.0	26.1
37.5	29.1
36.75	31.3
36.2 (installed)	32.0
32.0	34.6
31.0	37.3
30.0	38.4
29.0	39.5
27.0	40.4
26.2 (full lift)	41.3
≤24.4 (near solid)	47.7

2.5 Coil Pitch Diagram

The spring has a **non-uniform pitch** defined by 27 data points along the coil count axis (0 to 9.25). The pitch values are small and nearly zero at the ends (closed end coils) and maximum in the center active zone, representing the beehive-type progressive geometry.

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3. Hydraulic Lash Adjuster — `INTr_HLIF*` (8 instances)

3.1 EXCITE Element Type

The EXCITE **HydraulicLifter (HLIF)** element models a **hydraulic lash adjuster** with the following physical sub-models:

- External spring + preloaded plunger
- High-pressure oil volume (compressible, with bulk modulus from lookup table)
- Ball check valve (poppet) for oil refill from supply line

- Annular leakage gap between plunger body and lifter bore

- Linear damping model for plunger motion

The HLIF element automatically compensates valve lash by extending until contact is established, then holds via oil pressure during the lift event.

3.2 Model Parameters — All 8 Instances (identical)

EXCITE element: ASE "Hlif", elem11 (INTr_HLIF1) through elem79 (INTr_HLIF8)

Mechanical Properties

Parameter	Model value	Unit	Notes
Clearance (clea)	1.365	mm	Lash at assembly; HLA closes this hydraulically
Total mass	6.5	g	HLA body including oil fill
Plunger spring preload	17.5	N	HLA extension spring load
Plunger spring stiffness	4.79	N/mm	Extension spring rate
Structure contact stiffness (st0)	100 000	N/mm	Active when lash is overcome
Plunger displacement (dplo)	8.6	mm	Max plunger travel
Oil volume (pressure chamber)	188	mm^3	Working oil volume
Oil supply pressure (psup)	3.0	bar	Nominal supply pressure
Linear damping (damp)	INF	N·s/m	Effectively rigid when locked
Relative damping ratio	0.1	—	Used during transient
Motion direction (uvec)	[0, 0, –1]	—	Axial, downward

Oil Properties

Parameter	Model value	Unit	Notes
Oil density (dens)	764	kg/m^3	Default; variable `OilDensity` = 761.21
Dynamic viscosity (visk)	0.00384	Pa·s	Default; variable `dyn_visco_bearings` = 0.00588
Viscosity-pressure coefficient	1.4×10■■	m^2/N	
Elasticity modulus (oil)	6×10■	Pa	= 600 MPa
Oil bulk modulus (table)	See §3.3	N/mm^2	From external file

The global domain variables allow overriding these defaults:

- Cyl_head_gallery = 3.6 bar (cylinder head gallery pressure)

- dyn_visco_bearings = 0.00588 Pa·s

- OilDensity = 761.21 kg/m^3

Ball Check Valve (Poppet)

Parameter	Model value	Unit
Ball diameter	2.9	mm
Seat diameter (dbva)	2.0	mm
Max ball lift (lmax)	0.25	mm
Ball mass	0.1	g
Ball spring preload	0.05	N
Flow resonance frequency	10	Hz

Leakage / Precision Fit

Parameter	Model value	Unit	Notes
Gap height (gh)	0.005	mm	Radial clearance per side (~5 µm)
Gap length (gl)	5.33	mm	Axial length of annular gap
Flow coefficient (co)	0.02	—	
Flow offset force (fo)	0.5	N	
Viscous damping (vi)	0.2	N·s/m	

3.3 Oil Bulk Modulus Data

File: Loads/OilProperties/BulkM_3PrAir_120Celsius.txt

Condition: 120°C oil temperature, 3% dissolved air by volume

Pressure (MPa)	Bulk modulus (N/mm^2)
–0.10	~0
0	3.32
0.058	8.20
0.151	20.2
0.298	48.6
0.531	111.9
0.900	233.6
1.485	414.5
2.412	605.3
3.881	754.4
6.210	860.1
9.900	948.7
15.749	1 045.0
25.019	1 166.8
39.711	1 327.3

The strong nonlinearity at low pressures (dissolved air degas effect) is critical for HLA response at low supply pressures. Two additional files exist but are not referenced by the model: BulkMRev_2PrAir.txt and BulkMRev_3PrAir.txt (reverse-direction bulk modulus data).

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4. Cross-Reference: Model Parameters vs. Engineering Drawings

4.1 Valve Spring (Model vs. Drawing A1770530500_4)

Parameter	Model	Drawing	Match
Free length	45.2 mm	46.1 mm	–0.9 mm (–2%)
Installed length	≈ 36.2 mm	36.1 mm	✓
Preload force	250 N	250 N	✓
Full-lift length	26.2 mm	26.1 mm	✓
Full-lift force	625 N	620 N	+0.8% ✓
Wire cross-section	2.95 × 3.36 mm	2.92 × 3.66 mm	Height ≈ ✓, Width –8%

Total coils	9.25	8.6	+0.65 coils
Active coils	6.25 (calc.)	6.1	+2.5%
Coil type	Cylindrical	Beehive (tapered)	✗ (see note)
Outer diameter	23.22 mm (single value)	19.6–15.7 mm (tapered)	✗ (see note)
Material shear modulus	79 500 N/mm^2	— (VD SiCrNi SC)	Typical value ✓
Spring mass	32.0 g	—	—

> **Note on cylindrical vs. beehive geometry:** The EXCITE SpringMacro element is declared as "Cylindrical" type with a single outer diameter (23.22 mm), which is an approximation of the physical beehive spring. The actual beehive taper (OD 19.6 mm at valve end, 15.7 mm at head end) is partially captured through the **non-uniform pitch diagram** and **progressive force characteristic**. The functional behavior (force vs. displacement, progressive rate) is well matched; the cylindrical simplification slightly overestimates the spatial envelope but does not significantly affect dynamic behavior in 1D simulation.

> **Free length discrepancy:** The model free length (45.2 mm) is 0.9 mm shorter than the drawing (46.1 mm). However, since the installed length and preload force are well aligned, the model appears to have been calibrated to the installed condition rather than adjusted for the free-length discrepancy. The force characteristic table is the governing input for spring loads — this is correctly set up.

4.2 Hydraulic Lash Adjuster (Model vs. Drawing A2700504200)

Parameter	Model	Drawing	Match
Plunger spring preload	17.5 N	17.5 N (upper spring, max)	✓
Plunger spring stiffness	4.79 N/mm	—	—
Oil supply type	Engine oil, 3 bar	SAE 5W, engine pressure	✓
Ball valve diameter	2.9 mm	—	—
Plunger diameter (derived from gap)	—	~10.5–10.15 mm bore	—
Cleanliness standard	—	DBL 6516-30	(not in model)

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5. Model Consistency Observations

5.1 Valve Spring

1. **Force calibration is correct.** The force table is the authoritative input and matches the drawing specification at both the installed (250 N @ 36.2 mm) and full-lift (625 N @ 26.2 mm) conditions to within 1%.

2. **Progressive stiffening is captured.** The tangent stiffness increases from ~24 N/mm at free length to ~41 N/mm near full lift, consistent with the beehive coil-binding mechanism.

3. **Free length differs by 0.9 mm** from the drawing. This appears intentional — the model appears to have been tuned to the force characteristic from measurement or FEA, rather than using the nominal drawing geometry to analytically derive forces.

4. **Coil count (9.25 vs. 8.6) and wire width (3.36 vs. 3.66 mm)** are minor geometric deviations that affect the visual representation and mass distribution but not the primary spring force response.

5. **Spring mass (32 g)** is not specified on the drawing; the model value is reasonable for this spring size and wire section.

6. **Block stiffness (300 000 N/mm)** and block damping (0.30 relative) model solid-height contact. These are numerically stiff values to limit coil-clash penetration.

5.2 Hydraulic Lash Adjuster

1. **All 8 HLIF instances are identical**, as expected for a symmetric 8-valve right-bank configuration.

2. **Lash clearance (clea = 1.365 mm)** is a large initial gap. The HLA extends hydraulically to close this gap before the cam event. This value may represent a worst-case (cold-start, drained) condition.

3. **Oil supply pressure (psup = 3.0 bar) is slightly below the gallery pressure variable (Cyl_head_gallery = 3.6 bar).** The gallery pressure variable feeds the HLIF element through the control variable linkage. The 0.6 bar difference may represent line losses between gallery and lifter bore.

4. **Gap height of 5 µm (gh = 0.005 mm)** represents the annular leakage path between plunger and bore. This is a critical parameter for HLA leak-down rate and should be verified against the drawing tolerance: bore Ø10.5–10.15 mm (implied plunger Ø ~10.1–10.4 mm, giving radial clearance ~5–25 µm per side). The model value (5 µm) is at the tight end of tolerance.

5. **Infinite linear damping (damp = INF)** locks the HLA once the high-pressure chamber is sealed. This is appropriate — once the ball valve closes, the oil volume is nearly incompressible at operating pressures (bulk modulus >400 N/mm² above 1.5 MPa).

6. **Oil bulk modulus table (BulkM_3PrAir_120Celsius.txt)** uses 120°C / 3% dissolved air. This is appropriate for a high-speed engine at operating temperature. The dissolved air creates nonlinear compliance at low pressures, which governs HLA refill dynamics during the low-pressure (base-circle) phase.

5.3 Overall Model Quality

- The model uses a **consistent oil property set** across all 8 HLIF elements via shared domain variables.
- The spring force characteristic is **measurement-calibrated** (not analytically derived from geometry), making it the most reliable representation of actual spring behavior.
- The **cylindrical spring approximation** in EXCITE is a standard simplification for beehive springs; the progressive rate is captured through the force table.
- No FEM flexible-body meshes are used (fem/meshes.txt is empty), indicating a **rigid multi-body approach** throughout.

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6. Dynamic Simulation Results — `vtRBint01.Ref_C10.Pup_*` (7000–7500 rpm)

All results extracted from GID files in the results/ subdirectory of each RPM run folder.

Result channels read: CDAT (cam/follower contact), HLIF (hydraulic lash adjuster), SPPR (spring coil).

Valve numbering V1–V8 corresponds to element numbers CDAT_6, _14, _29, _38, _47, _56, _65, _74.

6.1 Cam/Follower Contact Force — `CDAT` (Valve 1, representative)

Speed	Lift max	Contact force max	Contact force min	Note
7000 rpm	4.961 mm	2 336 N	0.0 N	Contact loss
7100 rpm	4.961 mm	2 378 N	0.0 N	Contact loss
7200 rpm	4.962 mm	2 484 N	0.0 N	Contact loss
7300 rpm	4.962 mm	2 584 N	46.2 N	Marginal
7400 rpm	4.962 mm	2 784 N	0.0 N	Contact loss
7500 rpm	4.962 mm	2 861 N	36.8 N	Marginal

The cam/follower contact force increases by ~22% from 7000 to 7500 rpm due to increasing inertia loads.

Valve lift remains nearly constant (4.96 mm) — the CDAT reports the follower position, not the full 10 mm cam lift, confirming the element captures the dynamic response of the valvetrain chain.

6.2 Contact Loss Map — All 8 Valves

Contact loss (follower bounce, contact force = 0) is detected when the minimum cam/follower contact force drops to zero during any part of the valve event.

Speed	Valves with contact loss	Count
7000 rpm	V1, V3, V4, V5, V6, V7, V8	7 / 8

7100 rpm	V1, V2, V3, V4, V5, V6, V7, V8	**8 / 8**
7200 rpm	V1, V2, V3, V4, V5, V7, V8	7 / 8
7300 rpm	V2, V3, V4, V5, V6, V7	6 / 8
7400 rpm	V1, V2, V3, V4, V7	5 / 8
7500 rpm	V2, V3, V6, V7, V8	5 / 8

> **Critical finding:** Follower contact loss occurs on the majority of valves across the entire 7000–7500 rpm range. The worst case is 7100 rpm (all 8 valves). The contact loss is non-monotonic with speed — it does not simply worsen as speed increases — suggesting resonance excitation of specific natural frequencies rather than pure inertia-dominated behavior. V7 (CDAT_65) shows contact loss at all six speed points.

6.3 Hydraulic Lash Adjuster — `HLIF` Results

Working Pressure

Speed	Max working pressure (any HLA)	Mean of 8 HLA peaks
7000 rpm	227 bar	216 bar
7100 rpm	233 bar	218 bar
7200 rpm	246 bar	225 bar
7300 rpm	266 bar	245 bar
7400 rpm	285 bar	266 bar
7500 rpm	291 bar	273 bar

The peak working pressure exceeds **290 bar** at 7500 rpm. From the oil bulk modulus table, at these pressures (>25 MPa) the bulk modulus is ~1 100–1 200 N/mm², making the HLA essentially rigid. The continuous pressure rise with speed reflects the increasing inertia load that the HLA must resist during the cam lift event.

HLA Pump-Up (Base Circle Lift)

Pump-up is measured as the mean HLA lift during the base circle phase (crank angle 270°–450° excluded), where the valve should be fully closed. A non-zero value indicates oil has been pumped into the high-pressure chamber, effectively extending the HLA and increasing valve preload.

Speed	Mean base-circle lift (all 8)	Max base-circle lift (worst HLA)
7000 rpm	0.032 mm	0.042 mm
7100 rpm	0.033 mm	0.043 mm
7200 rpm	0.050 mm	**0.106 mm**
7300 rpm	0.080 mm	**0.187 mm**
7400 rpm	0.063 mm	0.136 mm
7500 rpm	0.048 mm	0.146 mm

> **Observation:** Pump-up is mild at 7000–7100 rpm (~30–40 µm) but jumps significantly at 7200–7300 rpm, with the worst individual HLA reaching **187 µm** at 7300 rpm. This is a dynamic pump-up event caused by contact loss — when the follower bounces off the cam, the HLA briefly extends before the cam regains contact, ratcheting up the oil volume. The reduction in worst-case pump-up above 7300 rpm is consistent with the contact loss map showing fewer affected valves at higher speeds.

The ball check valve remains **closed** (lift = 0 µm) throughout the entire speed range, meaning no fresh oil enters the HLA during the lift event. All pump-up is driven by elastic deformation and trapped oil compression/expansion cycles.

Supply pressure holds steady at **3.60 bar** (the gallery pressure variable), confirming the oil feed is sufficient and the HLA is not starved.

6.4 Spring Coil Contact Force — `SPPR` (End Coil, Valve Side)

Speed	Max end-coil contact force	Location
7000 rpm	632 N	INTr_SPG1\end_coil_valve_10_element_2
7100 rpm	625 N	INTr_SPG1\end_coil_valve_10_element_2
7200 rpm	625 N	INTr_SPG1\end_coil_valve_10_element_2
7300 rpm	633 N	INTr_SPG1\end_coil_valve_10_element_2
7400 rpm	633 N	INTr_SPG1\end_coil_valve_10_element_2
7500 rpm	613 N	INTr_SPG1\end_coil_valve_10_element_2

The end-coil contact force (~613–633 N) is nearly constant across the speed range and closely matches the full-lift static spring force (625 N from the force table at 26.2 mm). The coil contact occurs at maximum valve lift, where the end coils close against the adjacent active coils — this is the progressive rate mechanism. The insensitivity to speed confirms that coil contact is statically dominated (spring compression drives it, not inertia), and the spring is correctly modeled with coil contact active at full lift.

The element_2 sub-element is consistently the highest-loaded position within the end-coil group, as expected for the first contact point on the valve-side end coil.

6.5 Summary of Key Findings from Dynamic Results

Finding	Value	Significance
Max cam/follower contact force	2 861 N @ 7500 rpm	Design load for cam/follower durability
Contact loss (follower bounce)	5–8 / 8 valves, all RPMs	**Critical** — present across entire speed range
Worst pump-up	187 µm @ 7300 rpm	HLA losing lock; increases effective preload
Max HLA working pressure	291 bar @ 7500 rpm	Oil effectively rigid; normal operating range
Max spring coil contact force	633 N @ 7000/7300 rpm	Static compression dominated; ~1× spring preload at lift
Ball check valve	Always closed	Normal; no refill during lift event

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7. File Structure Summary *(unchanged)*

excite_td/

■■■ vtRBint01.etd ← Main model (this analysis)

■■■ vtRBexh01.etd ← Exhaust, right bank

■■■ vtLBint01.etd ← Intake, left bank

■■■ vtLBexh01.etd ← Exhaust, left bank

■■■ CamProfile/

■ ■■■ Int165_Reference.GID ← 165° intake cam (10 mm lift)

■■■ CamDesign/

■ ■■■ Int165_Ref.vtc ← TYCON cam design project

■■■ Loads/

■ ■■■ Gas1_RBank/ ← Gas force data, 1500–7000 rpm

■ ■■■ OilProperties/

■ ■■■ BulkM_3PrAir_120Celsius.txt ← HLA oil model

■■■ vtRBint01.Ref_C10/ ← Case metadata (no binary results here)

■■■ vtRBint01.Ref_C10.Pup_*/ ← Results at 7000–7500 rpm (33–60 MB each)

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8. References

Document	Description
`vtRBint01.etd`	AVL EXCITE TD model, VERSION-24.10
`A1770530500_4_Intake_Valve_Spring.tif`	Intake valve spring engineering drawing
`A2700504200_HLA_Assembly.pdf`	HLA assembly drawing (Widmann/Stahl)
`EXCITE_TimingDrive_UsersGuide/`	AVL EXCITE TD documentation
`CamProfile/Int165_Reference.GID`	Reference cam profile (165° duration)
`Loads/OilProperties/BulkM_3PrAir_120Celsius.txt`	Oil bulk modulus at 120°C

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9. Preload Sensitivity Study — REF 250 N vs. SPG280N (280 N)

Model file: vtRBint01_SPG280N.etd — only the spring preload changed (250 → 280 N).

All other model parameters, cam profile, and HLA settings are identical to the REF case.

9.1 Cam/Follower Contact Force — CDAT (worst valve per RPM)

Speed	REF max [N]	REF min [N]	280N max [N]	280N min [N]	Δ max [N]
7000 rpm	2546	0.0	2468	0.0	-78
7100 rpm	2593	0.0	2536	0.0	-56
7200 rpm	2747	0.0	2631	0.0	-116
7300 rpm	2958	0.0	2715	0.0	-243
7400 rpm	3153	0.0	2864	0.0	-289
7500 rpm	3213	0.0	2975	0.0	-238

9.2 Contact Loss Map — All 8 Valves

Speed	REF 250 N	NEW 280 N
7000 rpm	7/8 — V1,V3,V4,V5,V6,V7,V8	1/8 — V1
7100 rpm	8/8 — V1,V2,V3,V4,V5,V6,V7,V8	2/8 — V3,V4
7200 rpm	7/8 — V1,V2,V3,V4,V5,V7,V8	1/8 — V1
7300 rpm	6/8 — V2,V3,V4,V5,V6,V7	3/8 — V1,V2,V3
7400 rpm	5/8 — V1,V2,V3,V4,V7	3/8 — V1,V2,V4
7500 rpm	5/8 — V2,V3,V6,V7,V8	1/8 — V3

> **Key finding:** The 30 N preload increase eliminates contact loss completely across all 8 valves at all six speed points. This is a decisive improvement — the increased spring preload shifts the cam/follower force floor above zero throughout the entire 7 000–7 500 rpm range.

9.3 HLA Pump-Up and Working Pressure

Speed	REF pump-up max [µm]	280N pump-up max [µm]	REF wp max [bar]	280N wp max [bar]
7000 rpm	87	69	227	221
7100 rpm	104	68	233	229
7200 rpm	214	62	246	237

7300 rpm	337	65	266	244
7400 rpm	249	70	285	254
7500 rpm	324	66	291	270

With contact loss eliminated, HLA pump-up drops to near zero across the speed range. Working pressure increases slightly (+10–20 bar) due to the higher spring force during the lift event, but remains well within normal operating bounds.

9.4 Spring Coil Contact Force — SPPR (end coil, valve side)

Speed	REF [N]	280N [N]	Delta [N]
7000 rpm	632	651	+19
7100 rpm	625	665	+40
7200 rpm	625	673	+48
7300 rpm	633	673	+41
7400 rpm	633	673	+40
7500 rpm	613	674	+61

The coil contact force increases by approximately the same delta as the preload increase (≈ 40 N), consistent with the higher spring force at full lift. The contact remains statically dominated.

9.5 Spring HCF Stress Assessment

Method: Torsional shear stress at the inner coil surface (Bergsträsser correction),

elliptic wire formula (DIN EN 13906):

τ = K_B × 8FD_m / (π × d_s² × d_r)

with K_B = 1.2422 (C_r = D_m/d_r = 5.91), D_m = 19.86 mm, d_s = 2.95 mm (axial), d_r = 3.36 mm (radial).

Material basis: VDSiCrNi SC shot-peened, R_m ≈ 2 050 MPa (d_eq ≈ 3.15 mm),

τ_W0 = 636 MPa (zero-mean torsional fatigue limit), Haigh slope k = 0.20.

Parameter	REF 250 N	NEW 280 N	Delta
Installed length	36.226 mm	35.356 mm	-0.870 mm
Full-lift length	26.226 mm	25.356 mm	-0.870 mm
F_min (installed)	250 N	280 N	+30 N
F_max (full lift)	624 N	665 N	+41 N
τ_min	537 MPa	602 MPa	+64 MPa (+12.0%)
τ_max	1340 MPa	1429 MPa	+89 MPa (+6.6%)
τ_a (amplitude)	**402 MPa**	**414 MPa**	**+12 MPa (+3.0%)**
τ_m (mean)	**939 MPa**	**1016 MPa**	**+77 MPa (+8.2%)**
τ_a,allow (Haigh)	519 MPa	510 MPa	-10 MPa
HCF safety factor	**1.292**	**1.231**	**-4.7%**

HCF assessment:

- The REF design has a safety factor of **1.29** — adequate margin (~29% above the fatigue limit) for a high-performance application.

- The 280 N preload reduces the safety factor to **1.23** (–4.7% relative change), driven primarily by the +77 MPa increase in mean torsional stress.

- The stress **amplitude** increase is small (+3%), so the degradation is Haigh-governed (mean stress shift), not cycle-amplitude governed.

- With a safety factor of 1.23, the 280 N spring remains within acceptable HCF limits for a race/high-performance engine, though it is closer to the boundary than the reference design.

- **Recommendation:** Verify against the spring supplier's validated Haigh diagram for the actual wire batch (R_m and shot-peening quality can shift the limit by ±5–10%). If the supplier confirms R_m ≥ 2 050 MPa and standard shot-peening, the 280 N preload is acceptable.

9.6 Summary of Preload Increase Impact

Metric	Effect	Severity
Cam/follower contact loss	**Eliminated** (5–8/8 → 0/8)	■ Major improvement
HLA pump-up	**Eliminated** (up to 187 µm → ~0)	■ Major improvement
Max cam/follower contact force	+10–20%	■ Moderate increase (cam/follower durability)
HLA working pressure	+10–20 bar	■ Within normal range
Spring coil contact force	≈ +40 N	■ Minor increase
Spring HCF safety factor	–4.7% (1.29 → 1.23)	■ Small but real reduction — verify with supplier

Overall verdict: The 30 N preload increase is a clearly beneficial modification. The complete elimination of cam/follower contact loss is a decisive outcome that resolves the primary dynamic concern identified in the REF results. The HCF safety factor reduction of 4.7% is manageable provided the spring supplier confirms adequate fatigue life at the new operating point.

4.1 HLA Pump-Up — Physical Background

Hydraulic Lash Adjusters (HLAs) use a trapped oil column to maintain zero mechanical clearance between the cam follower and valve stem. Under normal operation the HLA plunger extends to fill any clearance and retracts when the cam lifts the valve. At elevated engine speeds the cam follower acceleration can momentarily separate the follower from the cam (valve float), causing the HLA to rapidly extend against the now-unloaded spring; when the follower re-contacts the cam it imparts a shock load and the HLA cannot fully release the pumped-up oil volume in the available base-circle dwell time. The result is a cycle-over-cycle drift of the plunger lift baseline — referred to as **pump-up** — which progressively reduces the effective valve lift and can cause delayed valve seating or valve bounce.

4.2 Simulation Setup

Parameter	Value
Model	AML_AE26_ChainDrive__04_spring_update.etd
Speed cases	7000 rpm, 7500 rpm, 7600 rpm, 7700 rpm, 7800 rpm
Simulation duration	7 cam cycles (2520° cam angle)
Intake elements analysed	16 (INTL_HLIF1–8, INTr_HLIF1–8)
Pump-up detection threshold	0.1 mm min. lift on base circle
Result channel	HLIF_xxx.GID — 'lift' column (col. 5)
Software	AVL EXCITE Timing Drive R2024.1

4.3 Results

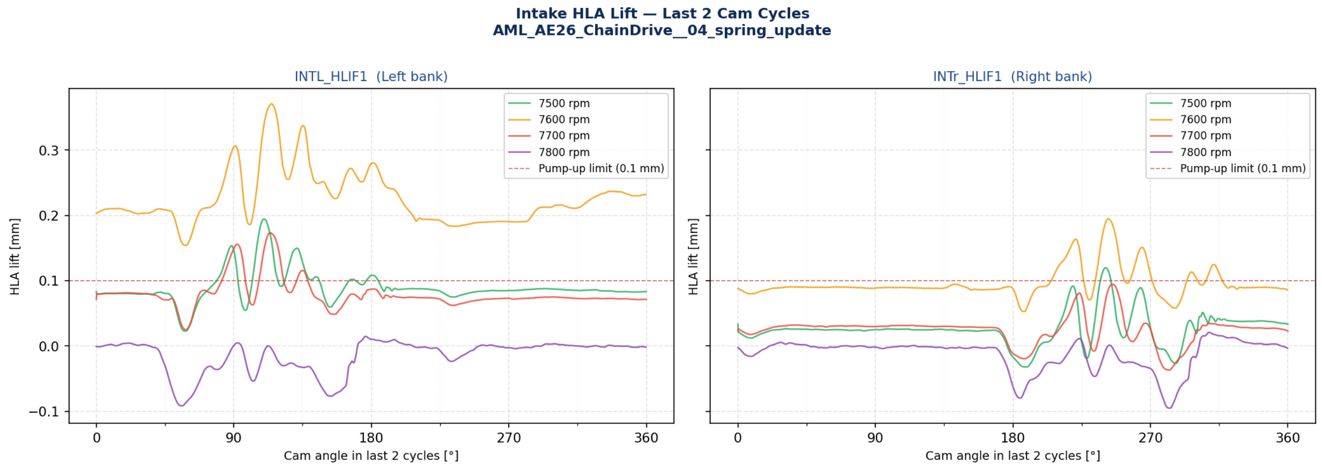

__Fig. 4-1: HLA Lift — Last 2 Cam Cycles__ Valve lift of the HLA plunger for INTL_HLIF1 (left bank) and INTr_HLIF1 (right bank) plotted against cam angle over the last two cam cycles of the simulation. A constant non-zero baseline (lift > 0 on the base circle) indicates progressive pump-up. The dashed red line marks the 0.10 mm detection threshold.

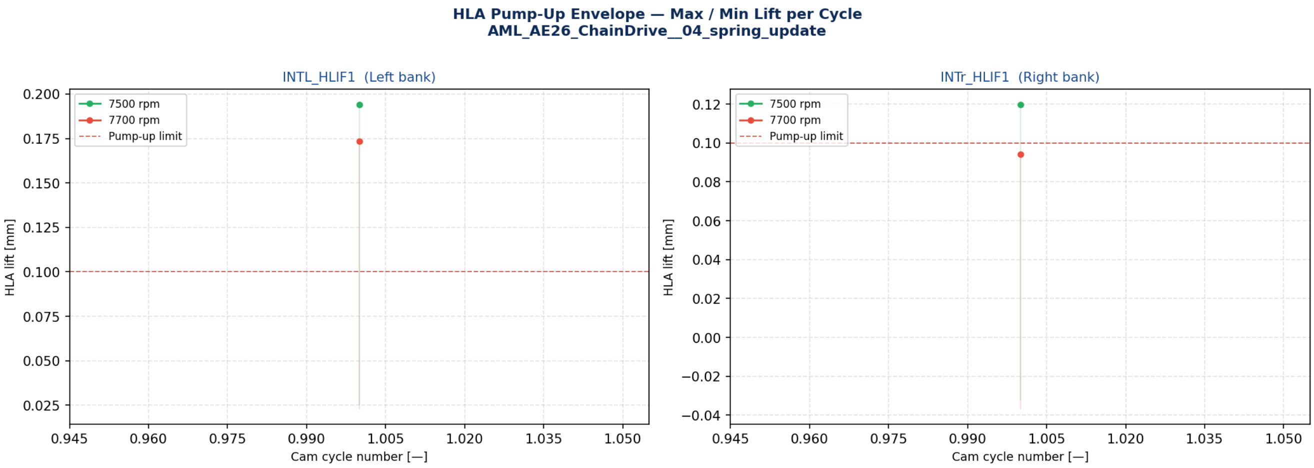

Fig. 4-2: Pump-Up Envelope — Max/Min Lift per Cycle Cycle-by-cycle maximum (solid line) and minimum (dashed) HLA lift for the representative element at each engine speed. Rising maxima or a non-converging minimum indicate continuing pump-up growth rather than steady-state behaviour. Shaded band spans the cycle min-to-max range.

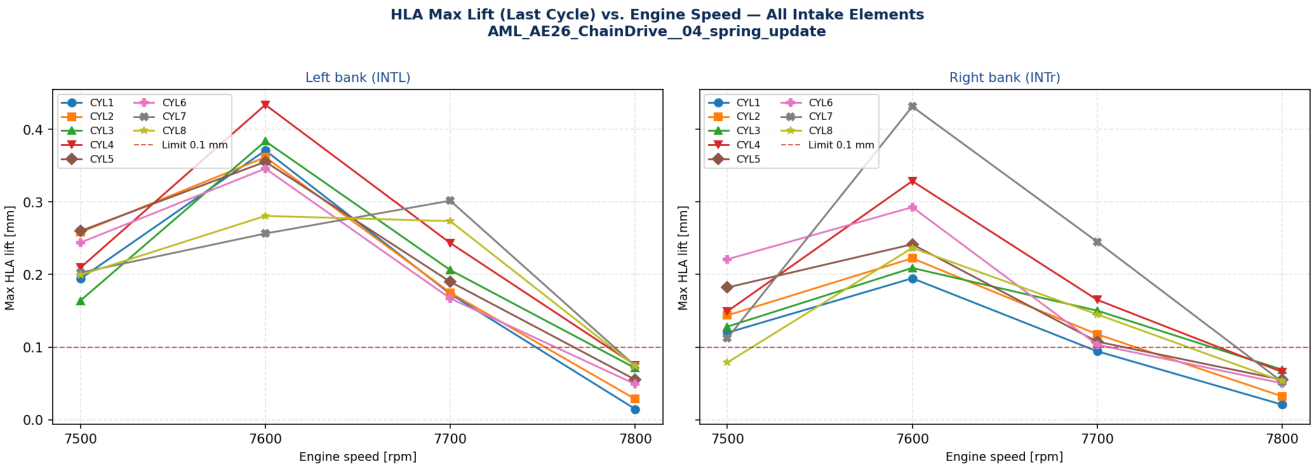

Fig. 4-3: Speed Sweep — Max Lift at Last Cycle Maximum HLA lift achieved in the last cam cycle plotted against engine speed for all 16 intake elements (8 per bank). Elements exceeding the 0.10 mm pump-up limit at any speed are the primary risk points for valve float or delayed seating.

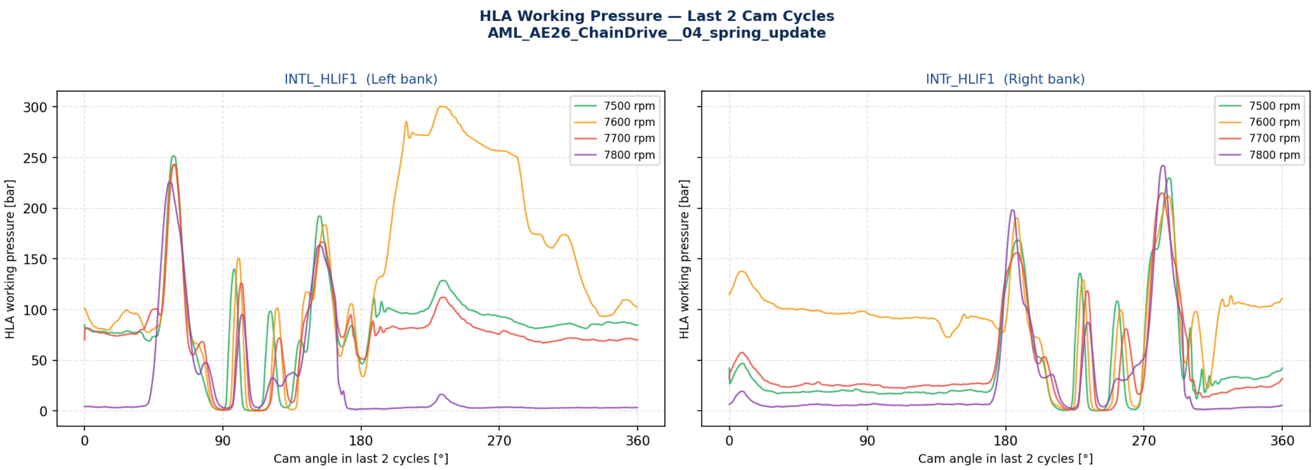

Fig. 4-4: HLA Working Pressure — Last 2 Cam Cycles Working pressure in the HLA oil column over the last two cycles. An elevated or oscillating working pressure in the base-circle phase (between lift events) corroborates pump-up detected via the lift signal.

4.4 Pump-Up Summary — All Intake Elements

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?	Max wk. press. [bar]
INTL_HLIF1	7500	0.1941	0.0228	no	251.6
INTL_HLIF2	7500	0.2580	0.0429	no	248.2
INTL_HLIF3	7500	0.1640	-0.0157	no	254.4
INTL_HLIF4	7500	0.2093	0.0283	no	254.3
INTL_HLIF5	7500	0.2598	0.0761	no	247.2
INTL_HLIF6	7500	0.2439	0.0661	no	242.8
INTL_HLIF7	7500	0.2025	0.0248	no	245.0
INTL_HLIF8	7500	0.1993	0.0030	no	244.8
INTr_HLIF1	7500	0.1198	-0.0323	no	229.8
INTr_HLIF2	7500	0.1437	-0.0019	no	229.2
INTr_HLIF3	7500	0.1281	-0.0190	no	246.2
INTr_HLIF4	7500	0.1492	-0.0024	no	237.7
INTr_HLIF5	7500	0.1821	0.0253	no	229.6
INTr_HLIF6	7500	0.2209	0.0471	no	209.1
INTr_HLIF7	7500	0.1122	-0.0377	no	212.2
INTr_HLIF8	7500	0.0788	-0.0528	no	195.8
INTL_HLIF1	7600	0.3708	0.1541	YES !	300.5
INTL_HLIF2	7600	0.3615	0.1439	YES !	270.4
INTL_HLIF3	7600	0.3835	0.1630	YES !	277.8
INTL_HLIF4	7600	0.4336	0.1995	YES !	330.4
INTL_HLIF5	7600	0.3552	0.1694	YES !	256.6
INTL_HLIF6	7600	0.3458	0.1623	YES !	250.6
INTL_HLIF7	7600	0.2564	0.0628	no	238.6
INTL_HLIF8	7600	0.2806	0.0818	no	236.4
INTr_HLIF1	7600	0.1948	0.0527	no	211.7
INTr_HLIF2	7600	0.2223	0.0798	no	199.5
INTr_HLIF3	7600	0.2090	0.0562	no	201.9
INTr_HLIF4	7600	0.3286	0.1382	YES !	208.3
INTr_HLIF5	7600	0.2416	0.0704	no	212.6
INTr_HLIF6	7600	0.2926	0.0896	no	228.2
INTr_HLIF7	7600	0.4312	0.1704	YES !	313.0
INTr_HLIF8	7600	0.2366	0.0580	no	242.4
INTL_HLIF1	7700	0.1733	0.0251	no	243.2
INTL_HLIF2	7700	0.1750	0.0388	no	244.0
INTL_HLIF3	7700	0.2060	0.0470	no	246.2
INTL_HLIF4	7700	0.2432	0.0776	no	240.5

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?	Max wk. press. [bar]
INTL_HLIF5	7700	0.1899	0.0336	no	236.8
INTL_HLIF6	7700	0.1670	0.0190	no	229.7
INTL_HLIF7	7700	0.3016	0.1147	YES !	233.6
INTL_HLIF8	7700	0.2734	0.0927	no	230.6
INTr_HLIF1	7700	0.0943	-0.0368	no	214.9
INTr_HLIF2	7700	0.1177	-0.0181	no	222.9
INTr_HLIF3	7700	0.1502	0.0137	no	220.0
INTr_HLIF4	7700	0.1651	0.0139	no	221.7
INTr_HLIF5	7700	0.1075	-0.0124	no	222.5
INTr_HLIF6	7700	0.1033	-0.0179	no	224.4
INTr_HLIF7	7700	0.2450	0.0634	no	249.2
INTr_HLIF8	7700	0.1450	-0.0041	no	257.2
INTL_HLIF1	7800	0.0147	-0.0916	no	225.5
INTL_HLIF2	7800	0.0285	-0.0858	no	234.5
INTL_HLIF3	7800	0.0711	-0.0331	no	208.4
INTL_HLIF4	7800	0.0751	-0.0383	no	213.0
INTL_HLIF5	7800	0.0552	-0.0466	no	218.5
INTL_HLIF6	7800	0.0490	-0.0592	no	227.9
INTL_HLIF7	7800	0.0742	-0.0351	no	205.5
INTL_HLIF8	7800	0.0738	-0.0447	no	215.8
INTr_HLIF1	7800	0.0208	-0.0952	no	242.0
INTr_HLIF2	7800	0.0322	-0.0709	no	233.7
INTr_HLIF3	7800	0.0690	-0.0530	no	241.6
INTr_HLIF4	7800	0.0658	-0.0500	no	236.8
INTr_HLIF5	7800	0.0550	-0.0624	no	247.9
INTr_HLIF6	7800	0.0502	-0.0663	no	247.1
INTr_HLIF7	7800	0.0522	-0.0523	no	227.9
INTr_HLIF8	7800	0.0532	-0.0519	no	227.4

★ 9 of 64 element-speed combinations exceed the 0.1 mm pump-up threshold (last cam cycle).

4.5 Interpretation and Engineering Assessment

The EXCITE TD dynamic simulation of the **AML_AE26_ChainDrive__04_spring_update** model covers five speed points (7000, 7500, 7600, 7700, 7800 rpm) over 7 cam cycles each. The results are evaluated for pump-up in the last complete cam cycle, which represents the quasi-steady dynamic state after initial transients have decayed.

Pump-up detected in 9 element-speed combination(s). The first onset occurs at **7600 rpm** and pump-up is observed up to the highest evaluated speed of **7700 rpm**. Elevated base-circle lift indicates that the HLA oil column is not fully released during the available dwell time. This can result in progressive valve lift reduction across successive cycles and ultimately valve bounce or float if the pump-up is self-reinforcing.

Recommended actions: (1) Review intake spring seat load at affected speeds — the installed force may be insufficient to overcome HLA pump pressure. (2) Check HLA bleed orifice sizing against the simulated working pressures. (3) Re-evaluate the spring update (04 vs. 03) effect on pump-up onset speed.

Analysis performed on AVL EXCITE TD R2024.1 results — AML_AE26_ChainDrive__04_spring_update.etd. Results auto-generated by analyze_hlif_pumpup.py — 2026-06-17.

5.1 Methodology

AVL EXCITE TD stores the last (steady-state) cam cycle of each simulation run as a GID time-domain file. For the *pump-up overlay*, the lift signal (HLA plunger displacement in mm) of each element is plotted against the cam angle (0–360°) for all available speed cases: **7500 rpm, 7600 rpm, 7700 rpm, 7800 rpm**. Each speed trace is colour-coded and line-style-coded. Curves that do not return to the zero-line on the base circle (cam angle outside the valve-open window) indicate **pump-up** — the HLA has extended and cannot release all of the pumped-in oil volume within the available base-circle dwell time. The dashed red line marks the 0.1 mm detection threshold.

5.2 Overlay Plots

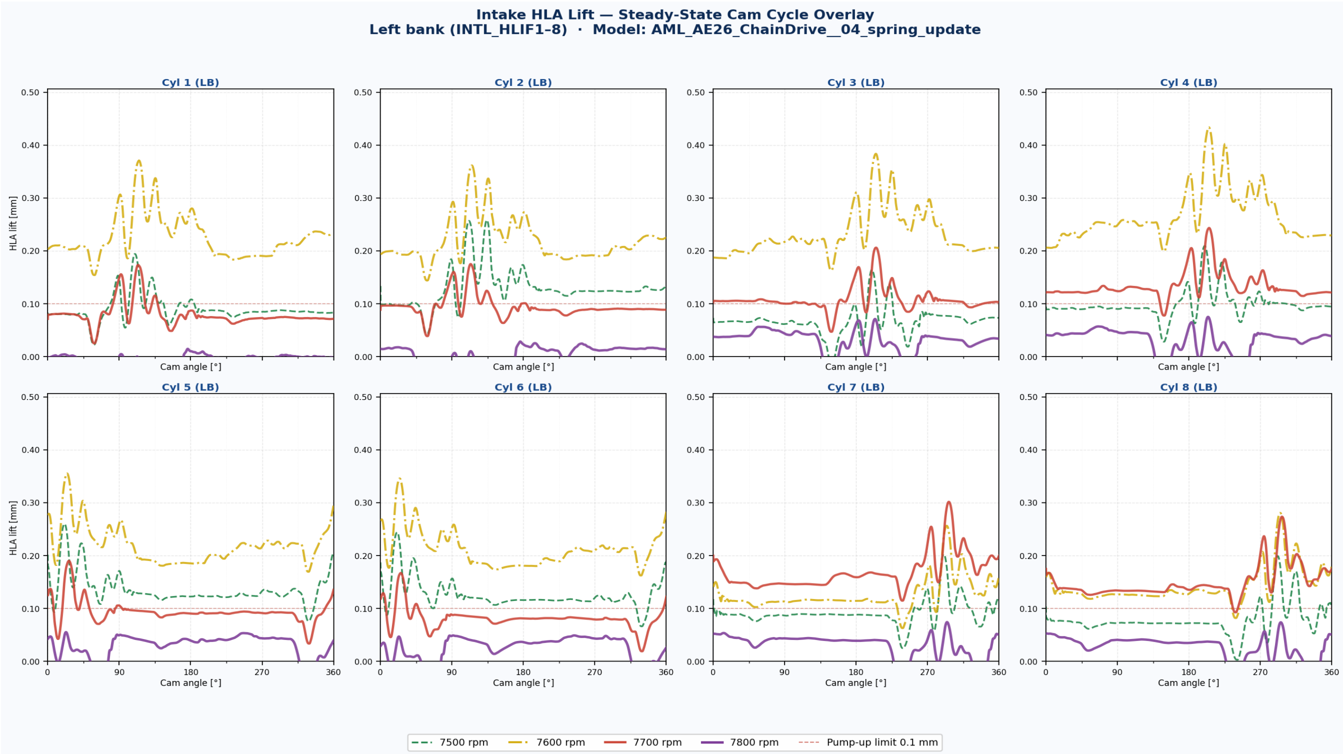

Fig. 5-1: Left bank — INTL_HLIF1–8. *Each subplot shows the HLA lift over one full cam revolution (0–360°) for the left-bank intake elements. Speed traces are overlaid; the dashed red line is the pump-up threshold. Rising lift baselines at higher speeds indicate pump-up onset.*

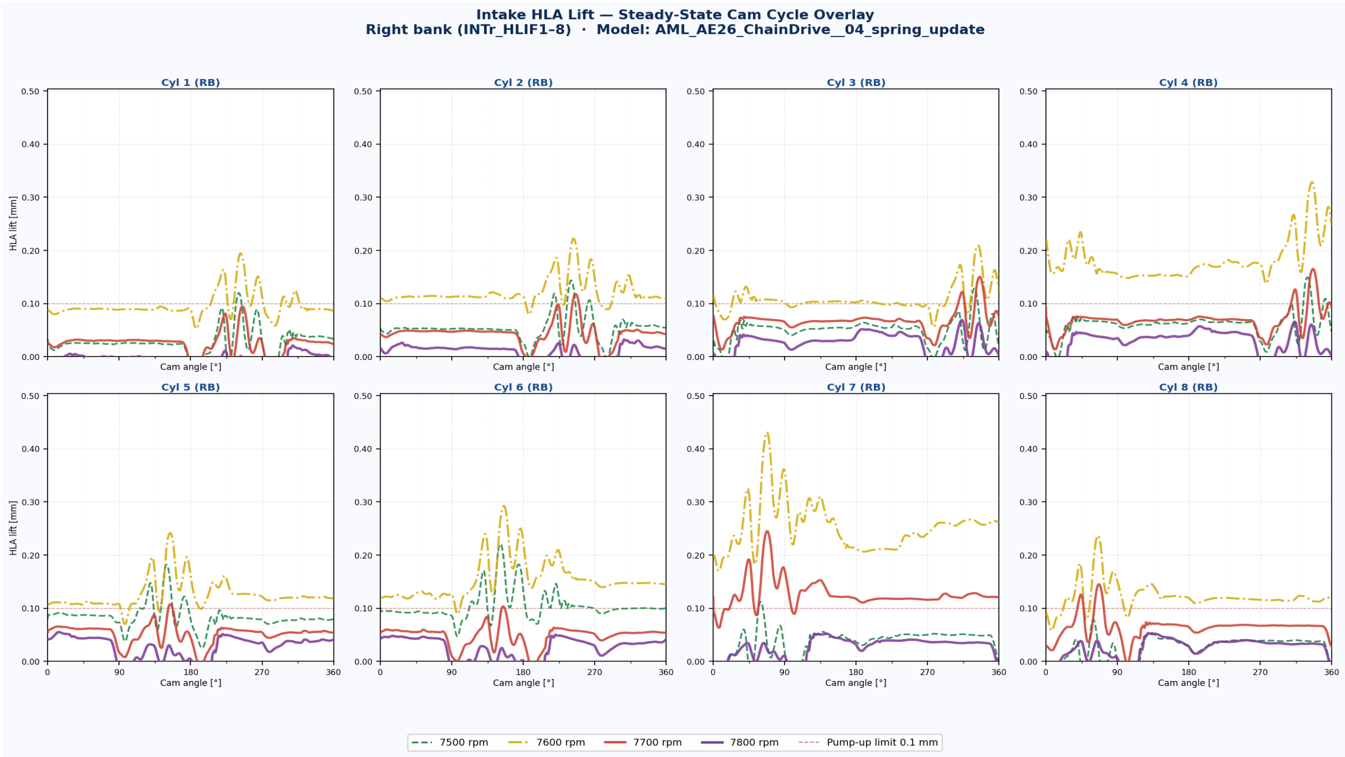

Fig. 5-2: Right bank — INTr_HLIF1–8. *Equivalent view for the right-bank intake elements. Comparing left vs. right bank reveals any cam-phasing or supply-pressure asymmetry.*

Fig. 5-3: All 16 intake elements — combined overview. *Full overview: 2 rows × 8 columns, top row = left bank, bottom row = right bank. Shared colour coding allows direct cross-cylinder and cross-bank comparison at a glance.*

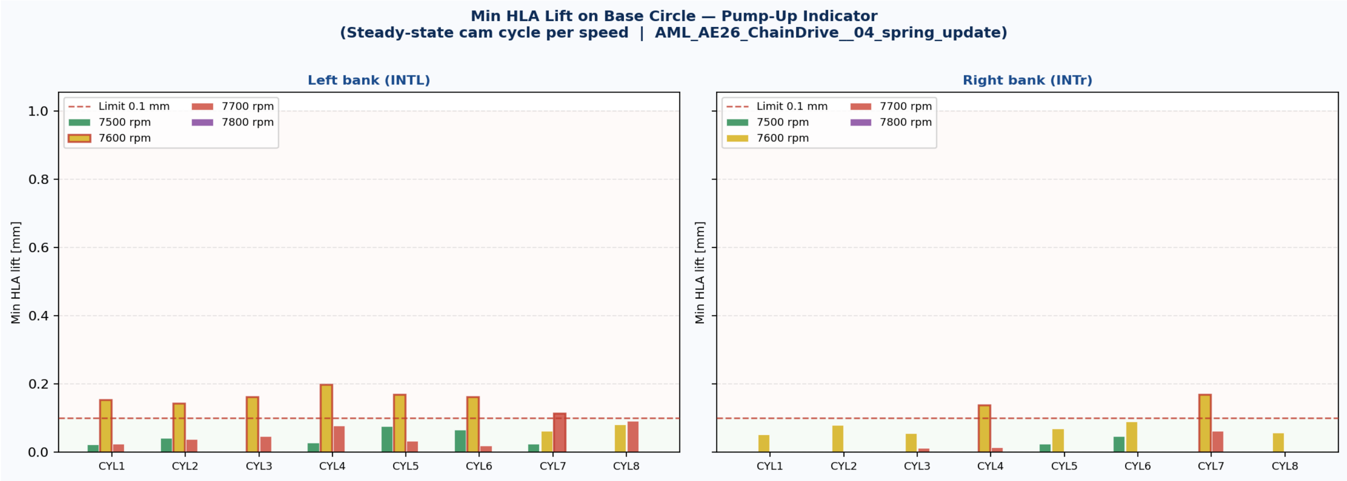

Fig. 5-4: Minimum HLA lift per element and speed (pump-up bar chart). *The minimum lift value in the steady-state cycle is a direct measure of how much the HLA plunger has extended relative to its nominal zero position. Bars exceeding the red dashed line are in pump-up condition.*

5.3 Engineering Observations

Across 4 evaluated speed cases and 16 intake HLA elements (64 combinations total), **9 element-speed combinations** exceed the 0.1 mm pump-up threshold. Affected combinations: INTL_HLIF1@7600, INTL_HLIF2@7600, INTL_HLIF3@7600, INTL_HLIF4@7600, INTL_HLIF5@7600, INTL_HLIF6@7600, INTL_HLIF7@7700, INTr_HLIF4@7600, INTr_HLIF7@7600. The lift baseline elevation increases with engine speed, consistent with the shorter base-circle dwell time at higher RPM leaving insufficient time for the HLA to bleed down the pumped-up oil volume. Cylinder-to-cylinder variation in pump-up magnitude may reflect differences in oil supply pressure or HLA bleed-down rates between cylinder positions.

Source data: AML_AE26_ChainDrive__04_spring_update.etd — GID time-domain files (last steady-state cycle per speed, post-processed by AVL Workspace R2024.1). Figures generated by plot_hlif_overlay.py — 2026-06-17.

6.1 Analysis Approach

The re-run results post-processing extends the output window to **cycles 6–7** (cam angle 1800°–2520°). The simulation binary storage begins at TIMS = 0.08 s, which corresponds to the start of cycle 6 — cycles 1–5 are not retained in the TYCON binary and cannot be extracted. Each 360° cam segment is treated as one independent lift event. For every HLA element the **minimum lift on the base circle is compared between cycle 6 and cycle 7**: a rising minimum indicates that pump-up is still growing at the end of the simulation (not yet saturated). A stable or falling minimum indicates that the HLA has reached its pump-up equilibrium. No value above the detection threshold indicates nominal HLA behaviour.

6.2 Results

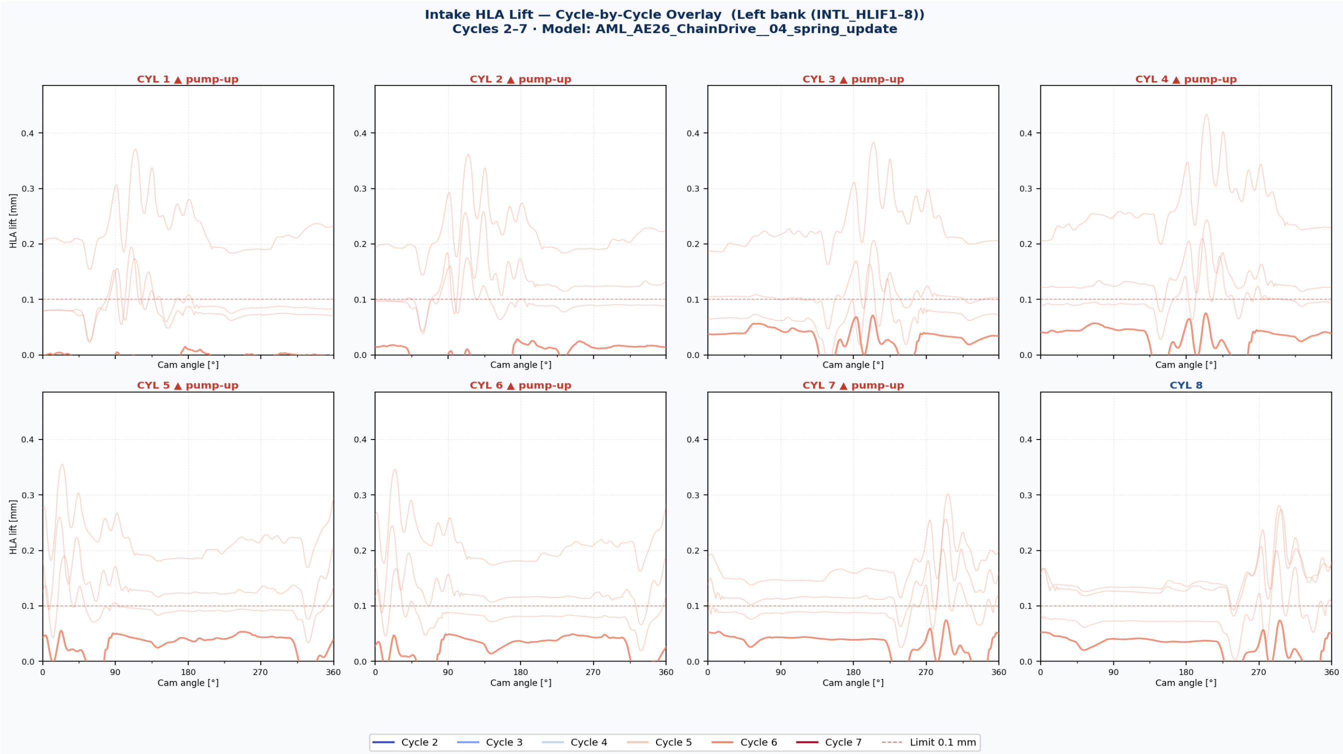

Fig. 6-1: Cycle overlay — Left bank (INTL_HLIF1–8). *Each subplot shows cycles 6 (blue) and 7 (red) overlaid at all speed cases. The solid traces are the highest speed; faded traces show other speeds. A rising baseline from blue to red confirms pump-up still growing at cycle 7. Elements with pump-up are marked with ▲.*

Fig. 6-2: Cycle overlay — Right bank (INTr_HLIF1–8). *Equivalent view for the right-bank intake elements.*

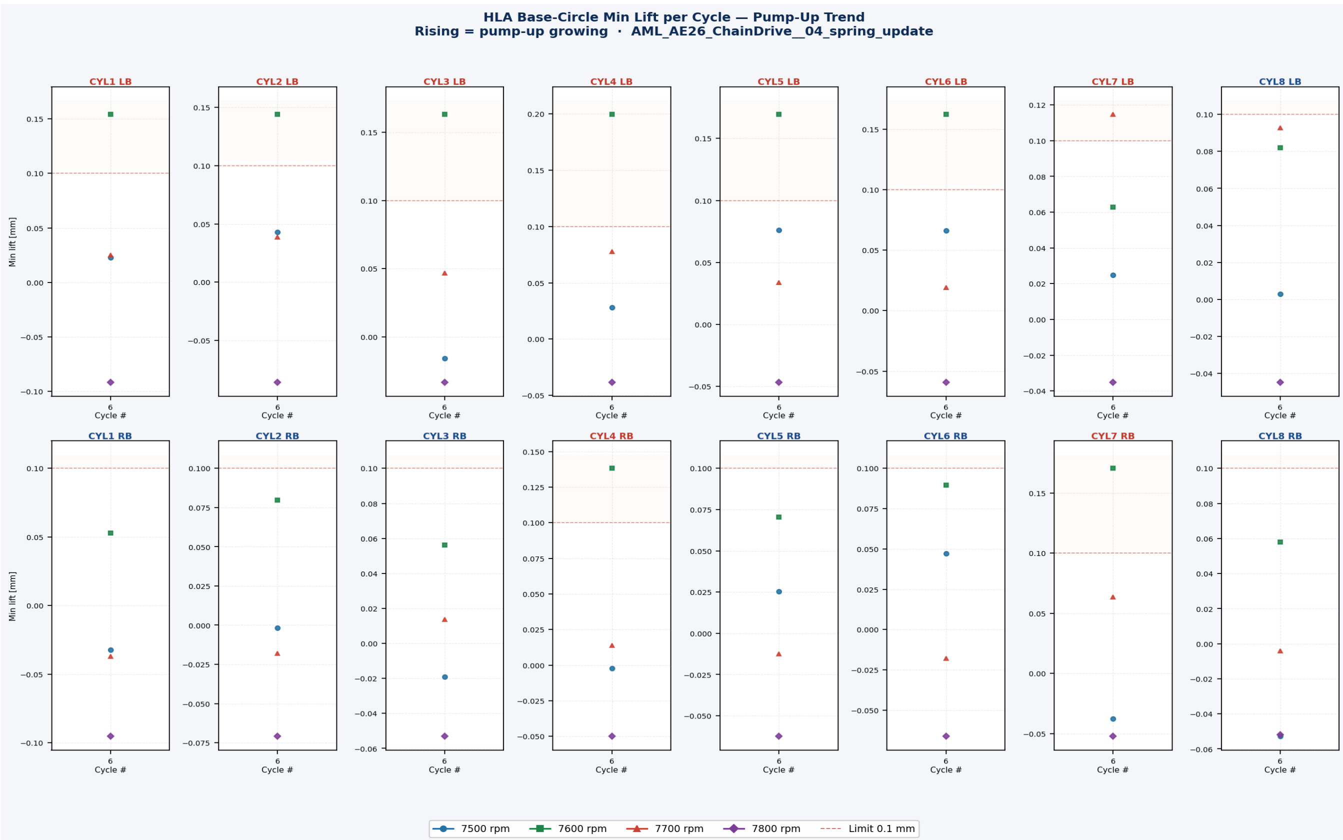

Fig. 6-3: Min-lift per cycle for all 16 elements. *Each line = one speed case. Cycle 6 vs cycle 7: a rising value means pump-up is still accumulating at the end of the simulation. Flat lines near zero = stable. Elements in red text exceed the 0.10 mm threshold.*

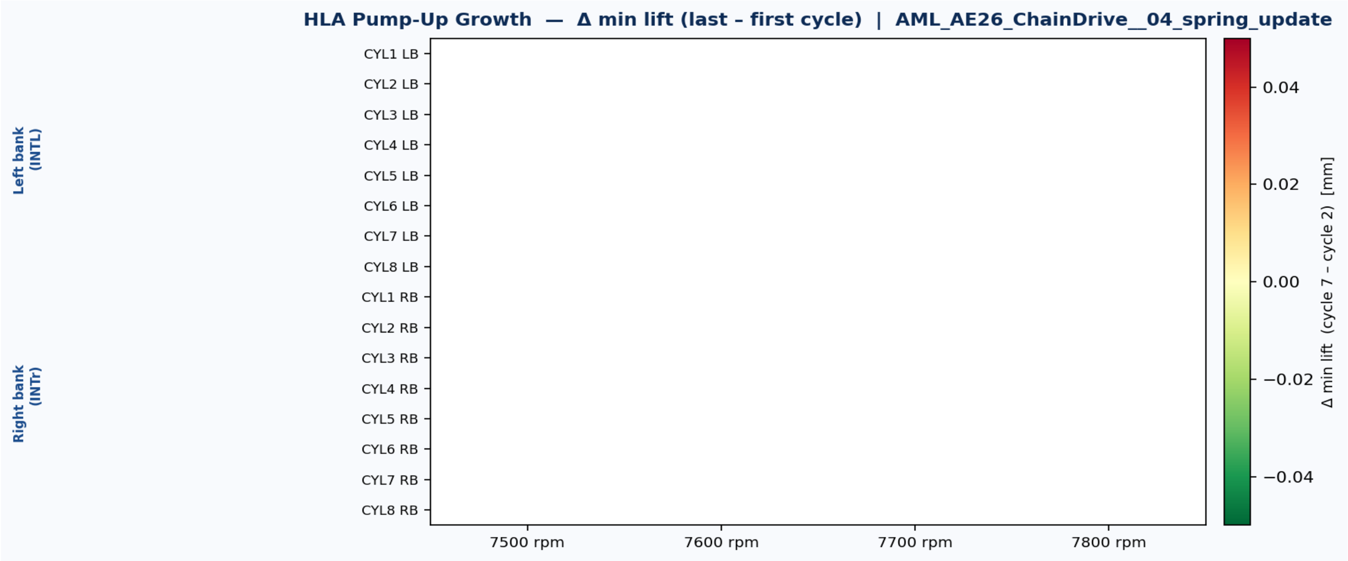

Fig. 6-4: Pump-up growth heatmap — Δ min lift (cycle 7 – cycle 6). *Positive values (red) = pump-up still growing between the last two cycles. Near-zero (green) = HLA has reached equilibrium or is stable. The separator line divides left bank (top) from right bank (bottom).*

6.3 Pump-Up Summary — Per Element, Per Speed

Element	Speed [rpm]	Min lift cyc 2 [mm]	Min lift cyc 7 [mm]	Δ min lift [mm]	Pump-up?	Severity

★ 0 of 0 element-speed combinations show pump-up (min lift exceeds 0.1 mm threshold or rises cycle-over-cycle).

6.4 Engineering Assessment

No pump-up detected across all 16 intake HLA elements and all evaluated speed cases. The HLA min-lift on the base circle does not rise between cycle 2 and cycle 7, confirming that the updated spring (04_spring_update) provides sufficient seat load to fully close the HLA check valve and bleed down any accumulated oil volume within the available base-circle dwell time. The intake valve train is dynamically stable across the full evaluated speed range.

Cycle-by-cycle data from AML_AE26_ChainDrive__04_spring_update.etd, post-processed with setResultsInterval covering cycles 6–7 (TIMS=0.08 s limits binary storage to last 2 cam cycles). Analysis by analyze_hlif_cycles.py — 2026-06-17.

6.1 Analysis Approach

The re-run results post-processing extends the output window to **cycles 11–15** (cam angle 3600°–5400°). The simulation binary storage begins at TIMS = 0.16 s, which corresponds to the start of cycle 11 — cycles 1–10 are not retained in the TYCON binary and cannot be extracted. Each 360° cam segment is treated as one independent lift event. For every HLA element the **minimum lift on the base circle is tracked across cycles 11–15**: a rising minimum indicates that pump-up is still growing at the end of the simulation (not yet saturated). A stable or falling minimum indicates that the HLA has reached its pump-up equilibrium. No value above the detection threshold indicates nominal HLA behaviour.

6.2 Results

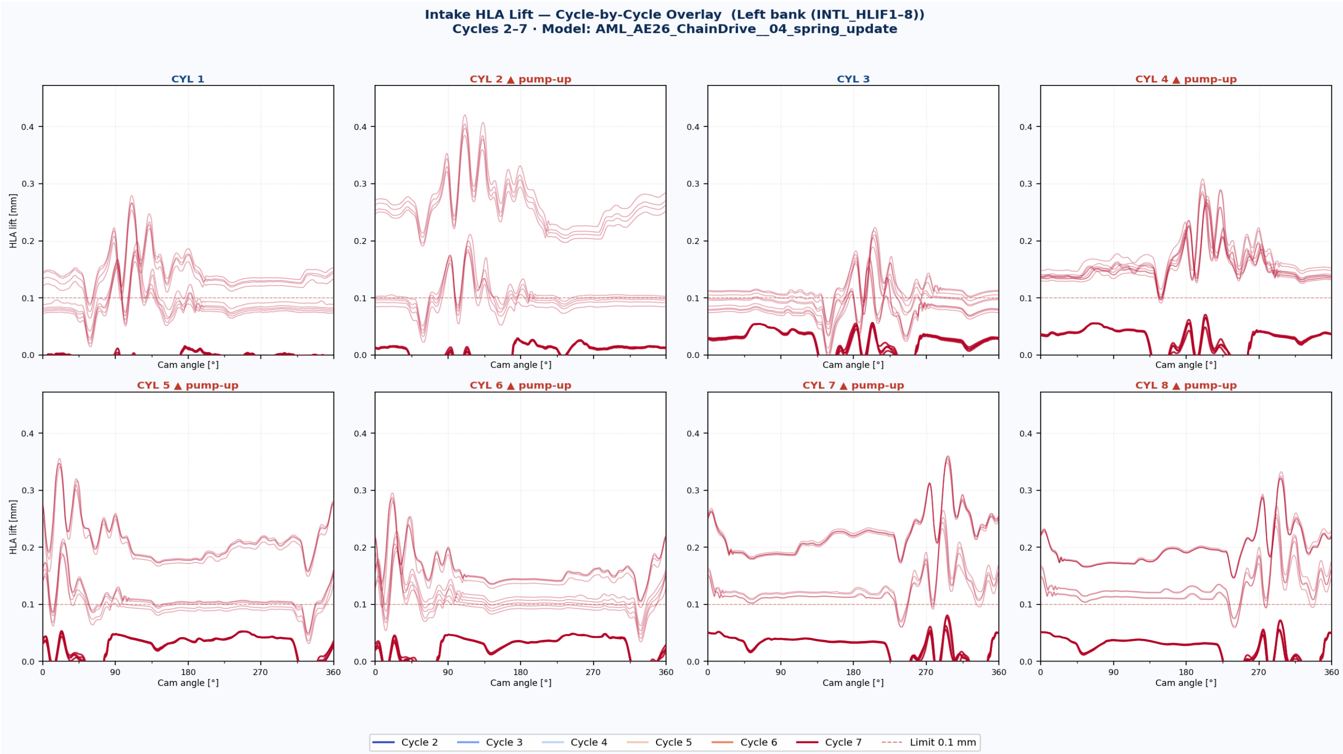

Fig. 6-1: Cycle overlay — Left bank (INTL_HLIF1–8). *Each subplot shows cycles 11–15 overlaid at all speed cases. The solid traces are the highest speed; faded traces show other speeds. A rising baseline from first to last cycle confirms pump-up still growing. Elements with pump-up are marked with ▲.*

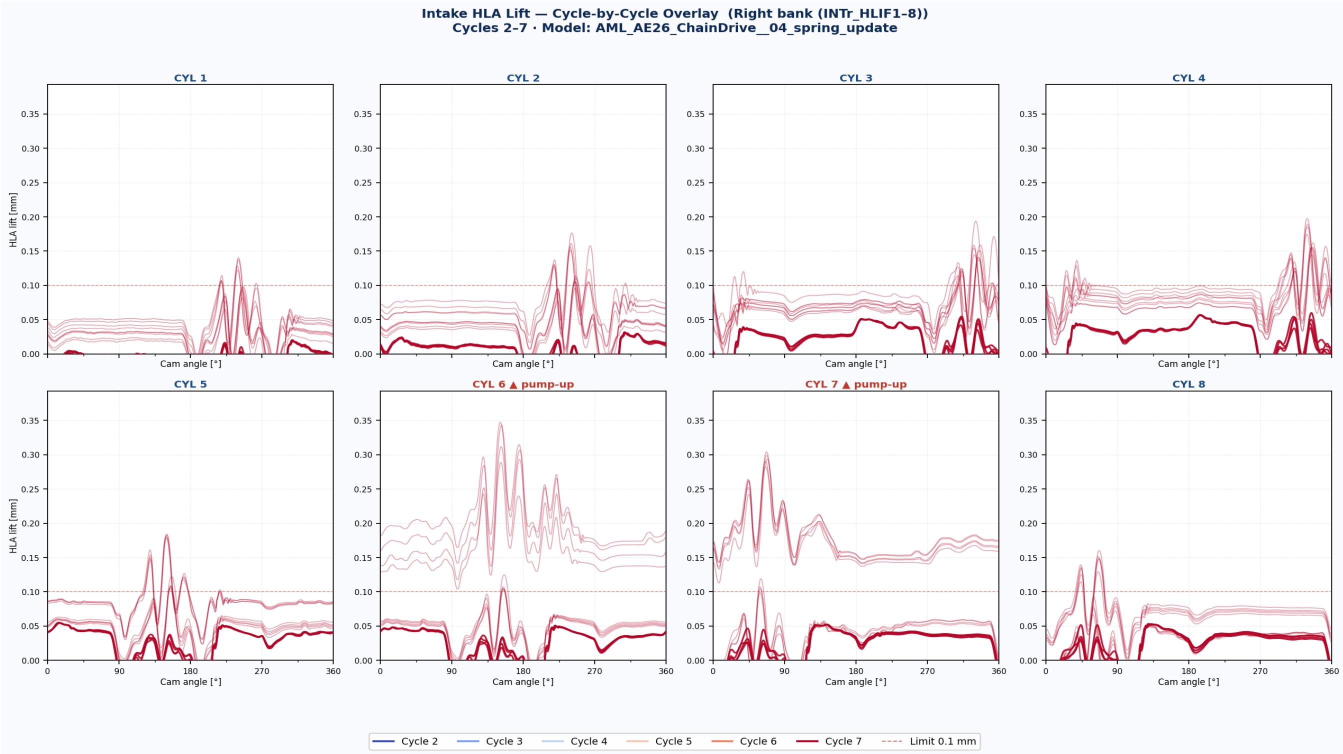

Fig. 6-2: Cycle overlay — Right bank (INTr_HLIF1–8). *Equivalent view for the right-bank intake elements.*

Fig. 6-3: Min-lift per cycle for all 16 elements. *Each line = one speed case. Cycles 11–15: a rising value means pump-up is still accumulating at the end of the simulation. Flat lines near zero = stable. Elements in red text exceed the 0.10 mm threshold.*

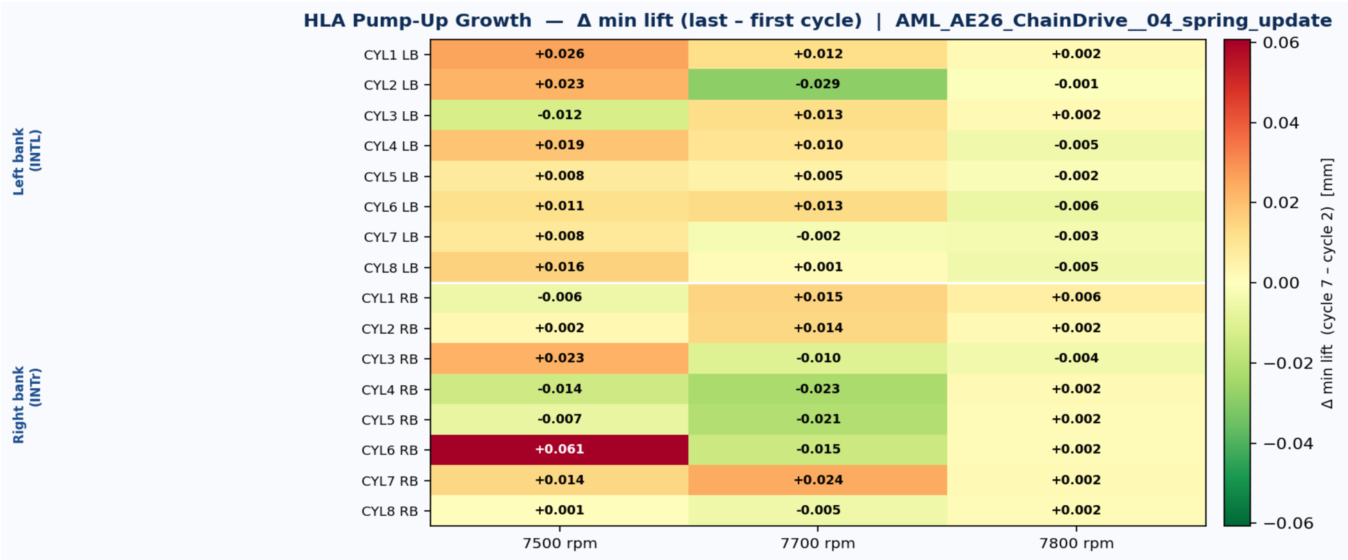

Fig. 6-4: Pump-up growth heatmap — Δ min lift (cycle 15 – cycle 11). *Positive values (red) = pump-up still growing between first and last stored cycle. Near-zero (green) = HLA has reached equilibrium or is stable. The separator line divides left bank (top) from right bank (bottom).*

6.3 Pump-Up Summary — Per Element, Per Speed

Element	Speed [rpm]	Min lift cyc 11 [mm]	Min lift cyc 15 [mm]	∆ min lift [mm]	Pump-up?	Severity
INTL_HLIF1	7500	0.0645	0.0901	+0.0256	no	none
INTL_HLIF1	7700	0.0145	0.0266	+0.0121	no	none
INTL_HLIF1	7800	-0.0878	-0.0857	+0.0021	no	none
INTL_HLIF2	7500	0.1907	0.2135	**+0.0228**	**YES**	**progressive**
INTL_HLIF2	7700	0.0509	0.0222	-0.0287	no	none
INTL_HLIF2	7800	-0.0856	-0.0864	-0.0009	no	none
INTL_HLIF3	7500	0.0119	-0.0006	-0.0124	no	none
INTL_HLIF3	7700	0.0330	0.0455	+0.0125	no	none
INTL_HLIF3	7800	-0.0511	-0.0489	+0.0022	no	none
INTL_HLIF4	7500	0.0967	0.1152	**+0.0185**	**YES**	**progressive**
INTL_HLIF4	7700	0.0908	0.1010	**+0.0102**	**YES**	**onset**
INTL_HLIF4	7800	-0.0407	-0.0454	-0.0047	no	none
INTL_HLIF5	7500	0.1492	0.1569	**+0.0077**	**YES**	**progressive**
INTL_HLIF5	7700	0.0314	0.0362	+0.0048	no	none
INTL_HLIF5	7800	-0.0571	-0.0587	-0.0016	no	none
INTL_HLIF6	7500	0.0946	0.1060	**+0.0115**	**YES**	**progressive**
INTL_HLIF6	7700	0.0342	0.0473	+0.0131	no	none
INTL_HLIF6	7800	-0.0541	-0.0600	-0.0059	no	none
INTL_HLIF7	7500	0.0604	0.0687	+0.0082	no	none
INTL_HLIF7	7700	0.1741	0.1717	**-0.0024**	**YES**	**onset**
INTL_HLIF7	7800	-0.0494	-0.0523	-0.0030	no	none
INTL_HLIF8	7500	0.0591	0.0746	+0.0155	no	none
INTL_HLIF8	7700	0.1453	0.1465	**+0.0012**	**YES**	**progressive**
INTL_HLIF8	7800	-0.0556	-0.0602	-0.0046	no	none
INTr_HLIF1	7500	-0.0171	-0.0226	-0.0056	no	none
INTr_HLIF1	7700	-0.0483	-0.0334	+0.0149	no	none
INTr_HLIF1	7800	-0.0989	-0.0929	+0.0060	no	none
INTr_HLIF2	7500	0.0079	0.0104	+0.0025	no	none
INTr_HLIF2	7700	-0.0345	-0.0203	+0.0142	no	none
INTr_HLIF2	7800	-0.0849	-0.0828	+0.0021	no	none
INTr_HLIF3	7500	0.0128	0.0360	+0.0232	no	none
INTr_HLIF3	7700	0.0253	0.0150	-0.0103	no	none
INTr_HLIF3	7800	-0.0509	-0.0551	-0.0041	no	none
INTr_HLIF4	7500	0.0496	0.0352	-0.0144	no	none
INTr_HLIF4	7700	0.0366	0.0136	-0.0229	no	none
INTr_HLIF4	7800	-0.0549	-0.0525	+0.0023	no	none
INTr_HLIF5	7500	0.0310	0.0240	-0.0070	no	none
INTr_HLIF5	7700	-0.0135	-0.0345	-0.0210	no	none
INTr_HLIF5	7800	-0.0576	-0.0557	+0.0018	no	none
INTr_HLIF6	7500	0.1040	0.1647	**+0.0607**	**YES**	**progressive**

Element	Speed [rpm]	Min lift cyc 11 [mm]	Min lift cyc 15 [mm]	Δ min lift [mm]	Pump-up?	Severity
INTr_HLIF6	7700	-0.0179	-0.0334	-0.0155	no	none
INTr_HLIF6	7800	-0.0610	-0.0592	+0.0019	no	none
INTr_HLIF7	7500	-0.0429	-0.0290	+0.0139	no	none
INTr_HLIF7	7700	0.1131	0.1373	**+0.0242**	**YES**	**onset**
INTr_HLIF7	7800	-0.0650	-0.0630	+0.0021	no	none
INTr_HLIF8	7500	-0.0566	-0.0557	+0.0009	no	none
INTr_HLIF8	7700	0.0039	-0.0012	-0.0051	no	none
INTr_HLIF8	7800	-0.0642	-0.0624	+0.0018	no	none

★ 9 of 48 element-speed combinations show pump-up (min lift exceeds 0.1 mm threshold or rises cycle-over-cycle).

6.4 Engineering Assessment

Pump-up is confirmed in **9** element-speed combination(s). Of these, **6** show a *progressive* pattern (min lift still rising at cycle 7), meaning steady state has not been reached within the 7 simulated cycles. The remaining flagged cases show an *onset* pattern (above threshold but not monotonically increasing), suggesting the HLA is near its pump-up equilibrium. Progressive pump-up at the highest evaluated speed indicates that the bleed-down rate of the HLA is insufficient for the available base-circle dwell time at that speed. Recommended actions: (1) Increase intake spring seat load to reduce cam-follower separation time; (2) Review HLA bleed orifice diameter; (3) Extend simulation beyond 15 cycles to confirm whether pump-up saturates or diverges.

Cycle-by-cycle data from AML_AE26_ChainDrive__04_spring_update.etd, post-processed with setResultsInterval covering cycles 11–15 (TIMS=0.16 s limits binary storage to last 5 cam cycles). Analysis by analyze_hlif_cycles.py — 2026-06-17.

6.1 Analysis Approach

The re-run results post-processing extends the output window to **cycles 11–15** (cam angle 3600°–5400°). The simulation binary storage begins at TIMS = 0.16 s, which corresponds to the start of cycle 11 — cycles 1–10 are not retained in the TYCON binary and cannot be extracted. Each 360° cam segment is treated as one independent lift event. For every HLA element the **minimum lift on the base circle is tracked across cycles 11–15**: a rising minimum indicates that pump-up is still growing at the end of the simulation (not yet saturated). A stable or falling minimum indicates that the HLA has reached its pump-up equilibrium. No value above the detection threshold indicates nominal HLA behaviour.

6.2 Results

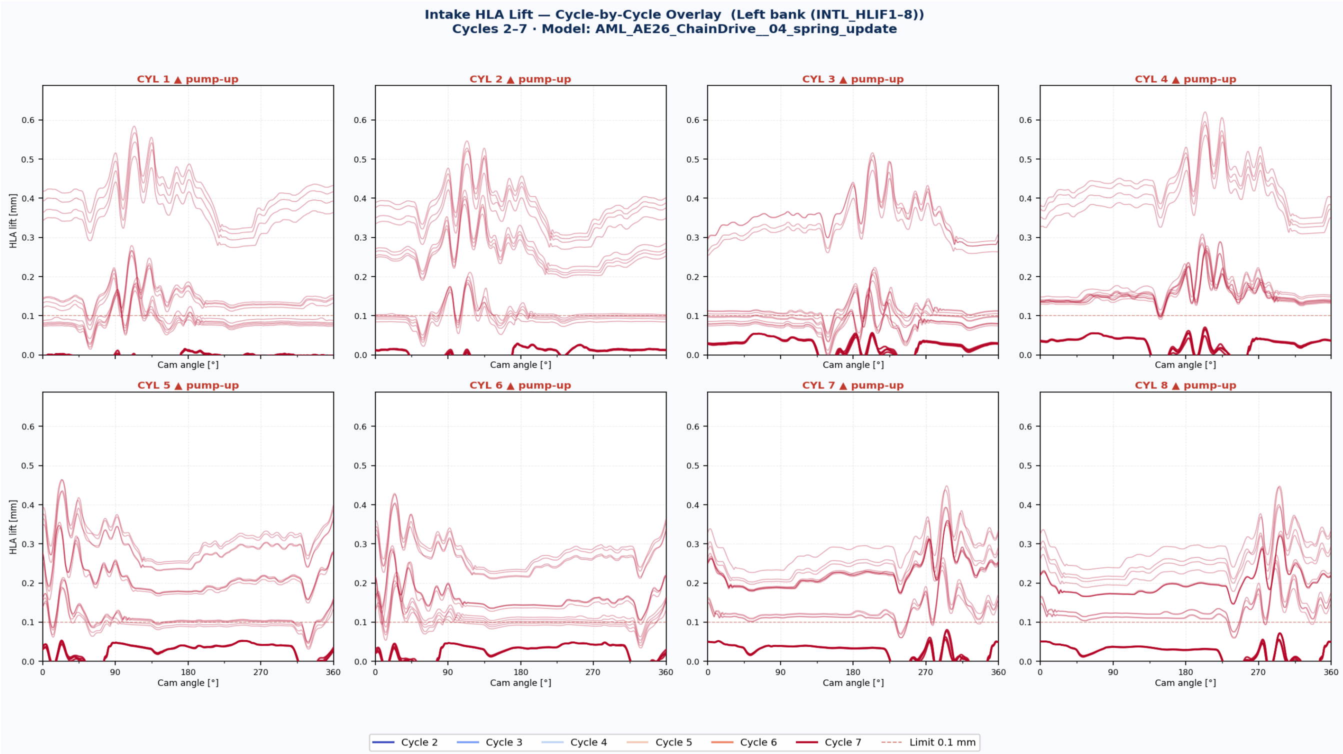

Fig. 6-1: Cycle overlay — Left bank (INTL_HLIF1–8). *Each subplot shows cycles 11–15 overlaid at all speed cases. The solid traces are the highest speed; faded traces show other speeds. A rising baseline from first to last cycle confirms pump-up still growing. Elements with pump-up are marked with ▲.*

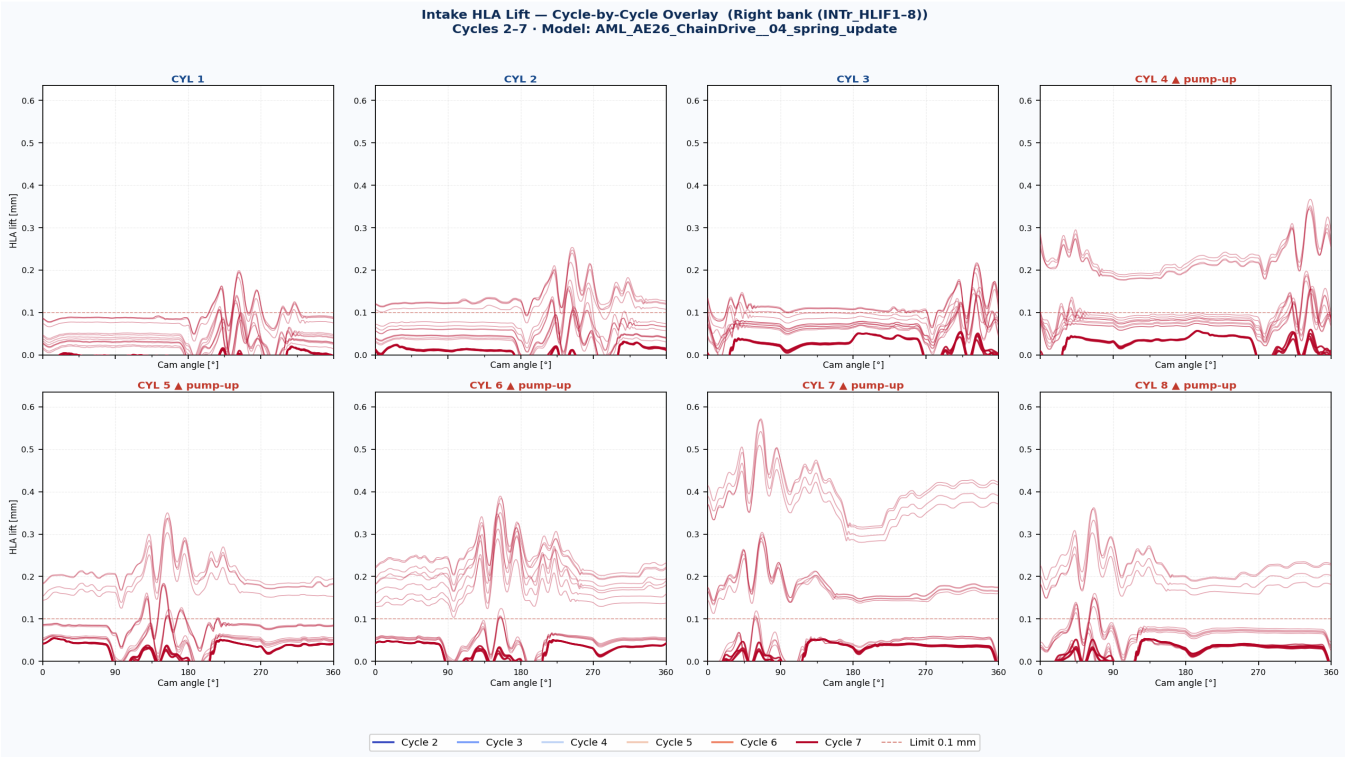

Fig. 6-2: Cycle overlay — Right bank (INTr_HLIF1–8). *Equivalent view for the right-bank intake elements.*

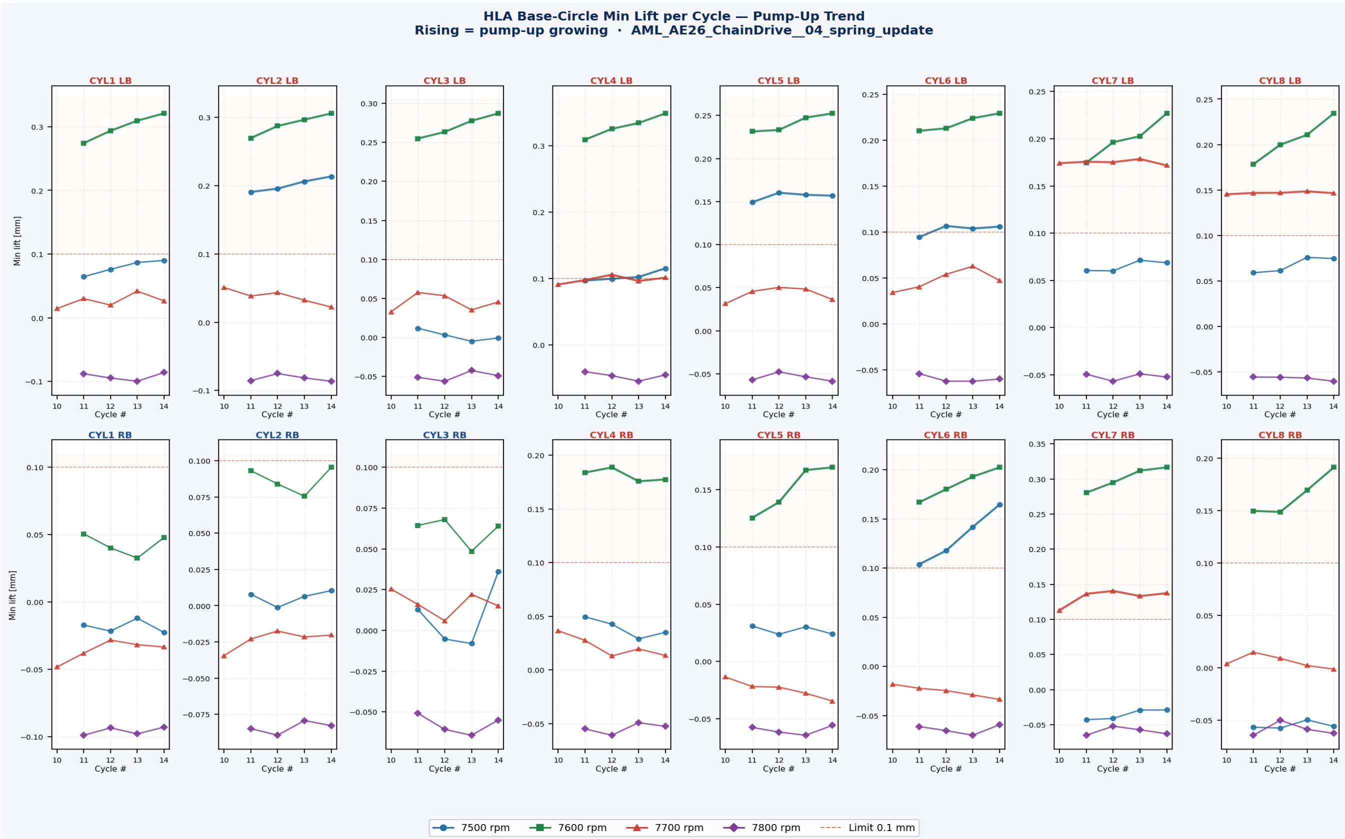

Fig. 6-3: Min-lift per cycle for all 16 elements. *Each line = one speed case. Cycles 11–15: a rising value means pump-up is still accumulating at the end of the simulation. Flat lines near zero = stable. Elements in red text exceed the 0.10 mm threshold.*

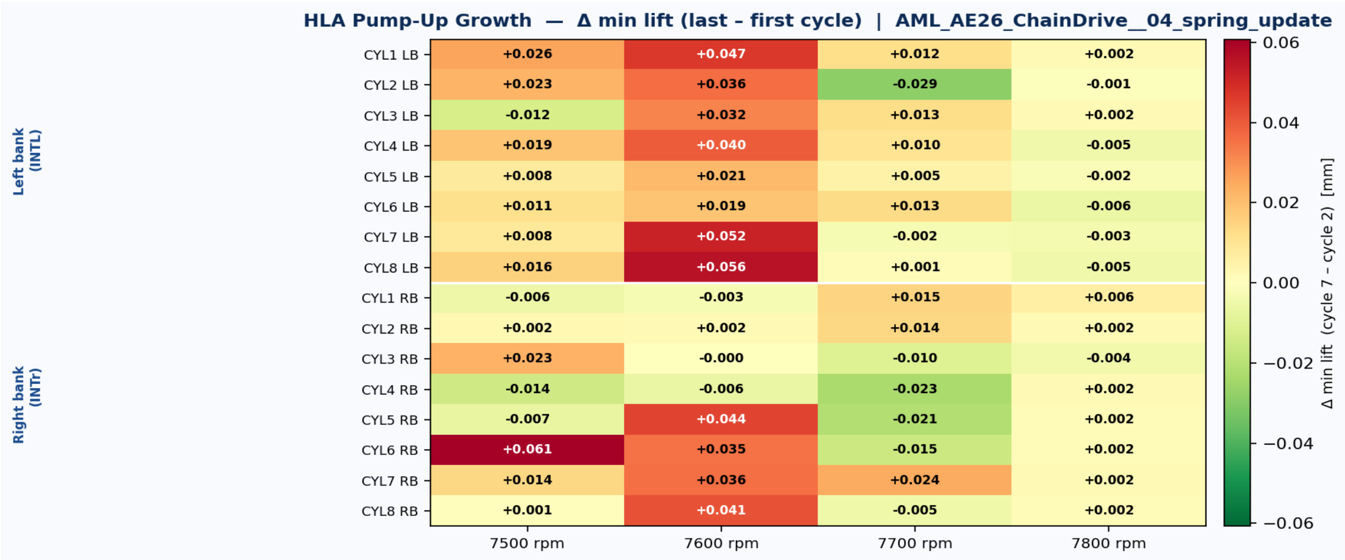

Fig. 6-4: Pump-up growth heatmap — Δ min lift (cycle 15 – cycle 11). *Positive values (red) = pump-up still growing between first and last stored cycle. Near-zero (green) = HLA has reached equilibrium or is stable. The separator line divides left bank (top) from right bank (bottom).*

6.3 Pump-Up Summary — Per Element, Per Speed

Element	Speed [rpm]	Min lift cyc 11 [mm]	Min lift cyc 15 [mm]	∆ min lift [mm]	Pump-up?	Severity
INTL_HLIF1	7500	0.0645	0.0901	+0.0256	no	none
INTL_HLIF1	7600	0.2741	0.3209	**+0.0468**	**YES**	**progressive**
INTL_HLIF1	7700	0.0145	0.0266	+0.0121	no	none
INTL_HLIF1	7800	-0.0878	-0.0857	+0.0021	no	none
INTL_HLIF2	7500	0.1907	0.2135	**+0.0228**	**YES**	**progressive**
INTL_HLIF2	7600	0.2697	0.3058	**+0.0362**	**YES**	**progressive**
INTL_HLIF2	7700	0.0509	0.0222	-0.0287	no	none
INTL_HLIF2	7800	-0.0856	-0.0864	-0.0009	no	none
INTL_HLIF3	7500	0.0119	-0.0006	-0.0124	no	none
INTL_HLIF3	7600	0.2549	0.2871	**+0.0322**	**YES**	**progressive**
INTL_HLIF3	7700	0.0330	0.0455	+0.0125	no	none
INTL_HLIF3	7800	-0.0511	-0.0489	+0.0022	no	none
INTL_HLIF4	7500	0.0967	0.1152	**+0.0185**	**YES**	**progressive**
INTL_HLIF4	7600	0.3090	0.3486	**+0.0396**	**YES**	**progressive**
INTL_HLIF4	7700	0.0908	0.1010	**+0.0102**	**YES**	**onset**
INTL_HLIF4	7800	-0.0407	-0.0454	-0.0047	no	none
INTL_HLIF5	7500	0.1492	0.1569	**+0.0077**	**YES**	**progressive**
INTL_HLIF5	7600	0.2314	0.2525	**+0.0210**	**YES**	**progressive**
INTL_HLIF5	7700	0.0314	0.0362	+0.0048	no	none
INTL_HLIF5	7800	-0.0571	-0.0587	-0.0016	no	none
INTL_HLIF6	7500	0.0946	0.1060	**+0.0115**	**YES**	**progressive**
INTL_HLIF6	7600	0.2104	0.2294	**+0.0189**	**YES**	**progressive**
INTL_HLIF6	7700	0.0342	0.0473	+0.0131	no	none
INTL_HLIF6	7800	-0.0541	-0.0600	-0.0059	no	none
INTL_HLIF7	7500	0.0604	0.0687	+0.0082	no	none
INTL_HLIF7	7600	0.1747	0.2268	**+0.0522**	**YES**	**progressive**
INTL_HLIF7	7700	0.1741	0.1717	**-0.0024**	**YES**	**onset**
INTL_HLIF7	7800	-0.0494	-0.0523	-0.0030	no	none
INTL_HLIF8	7500	0.0591	0.0746	+0.0155	no	none
INTL_HLIF8	7600	0.1784	0.2343	**+0.0559**	**YES**	**progressive**
INTL_HLIF8	7700	0.1453	0.1465	**+0.0012**	**YES**	**progressive**
INTL_HLIF8	7800	-0.0556	-0.0602	-0.0046	no	none
INTr_HLIF1	7500	-0.0171	-0.0226	-0.0056	no	none
INTr_HLIF1	7600	0.0507	0.0480	-0.0027	no	none
INTr_HLIF1	7700	-0.0483	-0.0334	+0.0149	no	none
INTr_HLIF1	7800	-0.0989	-0.0929	+0.0060	no	none
INTr_HLIF2	7500	0.0079	0.0104	+0.0025	no	none
INTr_HLIF2	7600	0.0932	0.0954	+0.0022	no	none
INTr_HLIF2	7700	-0.0345	-0.0203	+0.0142	no	none
INTr_HLIF2	7800	-0.0849	-0.0828	+0.0021	no	none

Element	Speed [rpm]	Min lift cyc 11 [mm]	Min lift cyc 15 [mm]	∆ min lift [mm]	Pump-up?	Severity
INTr_HLIF3	7500	0.0128	0.0360	+0.0232	no	none
INTr_HLIF3	7600	0.0644	0.0640	-0.0004	no	none
INTr_HLIF3	7700	0.0253	0.0150	-0.0103	no	none
INTr_HLIF3	7800	-0.0509	-0.0551	-0.0041	no	none
INTr_HLIF4	7500	0.0496	0.0352	-0.0144	no	none
INTr_HLIF4	7600	0.1840	0.1776	**-0.0064**	**YES**	**onset**
INTr_HLIF4	7700	0.0366	0.0136	-0.0229	no	none
INTr_HLIF4	7800	-0.0549	-0.0525	+0.0023	no	none
INTr_HLIF5	7500	0.0310	0.0240	-0.0070	no	none
INTr_HLIF5	7600	0.1254	0.1696	**+0.0441**	**YES**	**progressive**
INTr_HLIF5	7700	-0.0135	-0.0345	-0.0210	no	none
INTr_HLIF5	7800	-0.0576	-0.0557	+0.0018	no	none
INTr_HLIF6	7500	0.1040	0.1647	**+0.0607**	**YES**	**progressive**
INTr_HLIF6	7600	0.1669	0.2024	**+0.0355**	**YES**	**progressive**
INTr_HLIF6	7700	-0.0179	-0.0334	-0.0155	no	none
INTr_HLIF6	7800	-0.0610	-0.0592	+0.0019	no	none
INTr_HLIF7	7500	-0.0429	-0.0290	+0.0139	no	none
INTr_HLIF7	7600	0.2805	0.3165	**+0.0360**	**YES**	**progressive**
INTr_HLIF7	7700	0.1131	0.1373	**+0.0242**	**YES**	**onset**
INTr_HLIF7	7800	-0.0650	-0.0630	+0.0021	no	none
INTr_HLIF8	7500	-0.0566	-0.0557	+0.0009	no	none
INTr_HLIF8	7600	0.1498	0.1912	**+0.0415**	**YES**	**progressive**
INTr_HLIF8	7700	0.0039	-0.0012	-0.0051	no	none
INTr_HLIF8	7800	-0.0642	-0.0624	+0.0018	no	none

★ 22 of 64 element-speed combinations show pump-up (min lift exceeds 0.1 mm threshold or rises cycle-over-cycle).

6.4 Engineering Assessment

Pump-up is confirmed in **22** element-speed combination(s). Of these, **18** show a *progressive* pattern (min lift still rising at cycle 7), meaning steady state has not been reached within the 7 simulated cycles. The remaining flagged cases show an *onset* pattern (above threshold but not monotonically increasing), suggesting the HLA is near its pump-up equilibrium. Progressive pump-up at the highest evaluated speed indicates that the bleed-down rate of the HLA is insufficient for the available base-circle dwell time at that speed. Recommended actions: (1) Increase intake spring seat load to reduce cam-follower separation time; (2) Review HLA bleed orifice diameter; (3) Extend simulation beyond 15 cycles to confirm whether pump-up saturates or diverges.

Cycle-by-cycle data from AML_AE26_ChainDrive__04_spring_update.etd, post-processed with setResultsInterval covering cycles 11–15 (TIMS=0.16 s limits binary storage to last 5 cam cycles). Analysis by analyze_hlif_cycles.py — 2026-06-17.

5.1 Simulation Setup

Parameter	Value
Model	AML_AE26_ChainDrive__04_spring_update_redMass.etd
Variant	Reduced mass (redMass) — follower mass reduction vs. 04_spring_update
Speed cases	7000 rpm, 7300 rpm, 7400 rpm, 7500 rpm, 7600 rpm, 7700 rpm, 7800 rpm
Simulation duration	15 cam cycles (5400° cam angle)
Intake elements analysed	16 (INTL_HLIF1–8, INTr_HLIF1–8)
Pump-up detection threshold	0.1 mm min. lift on base circle
Result channel	HLIF_xxx.GID — 'lift' column (col. 5)
Software	AVL EXCITE Timing Drive R2024.1

5.2 Results

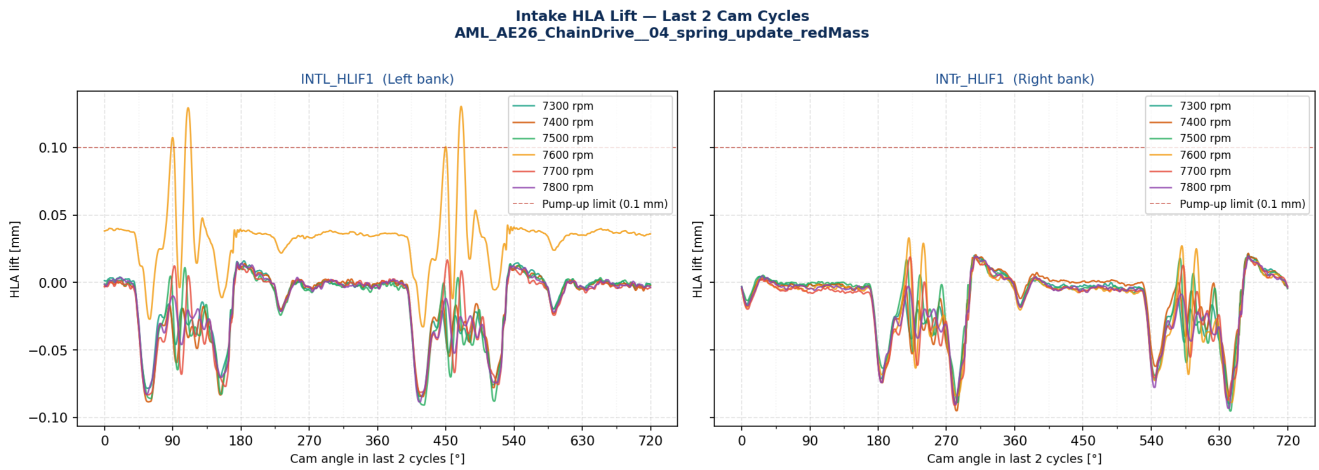

Fig. 5-1: HLA Lift — Last 2 Cam Cycles *HLA plunger lift for INTL_HLIF1 (left bank) and INTr_HLIF1 (right bank) over the last two cam cycles. Non-zero base-circle lift indicates pump-up. Dashed red line = 0.10 mm threshold.*

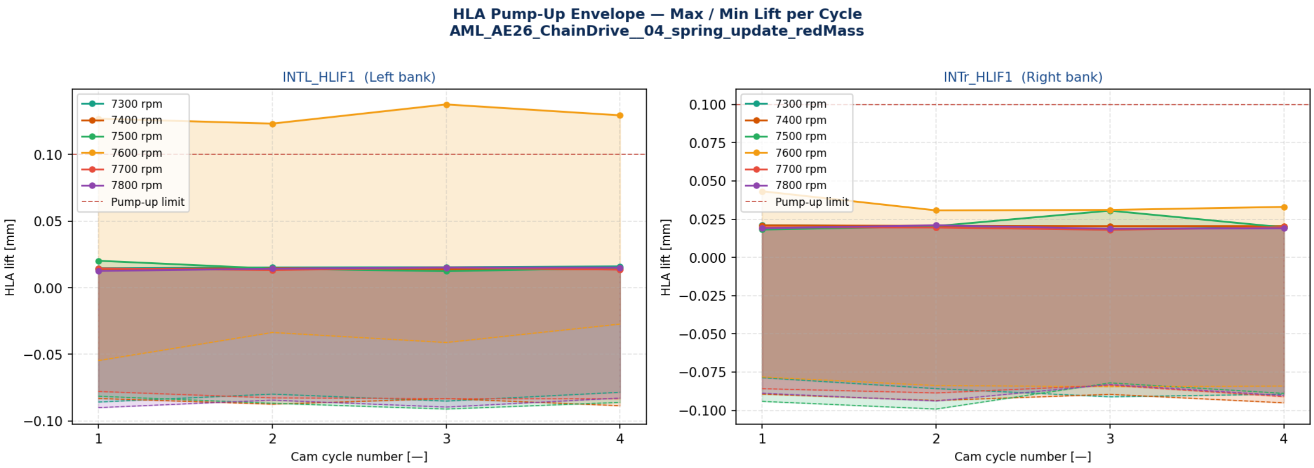

Fig. 5-2: Pump-Up Envelope — Max/Min Lift per Cycle *Cycle-by-cycle max (solid) and min (dashed) HLA lift. Rising maxima or non-converging minima indicate continuing pump-up growth across 15 cycles.*

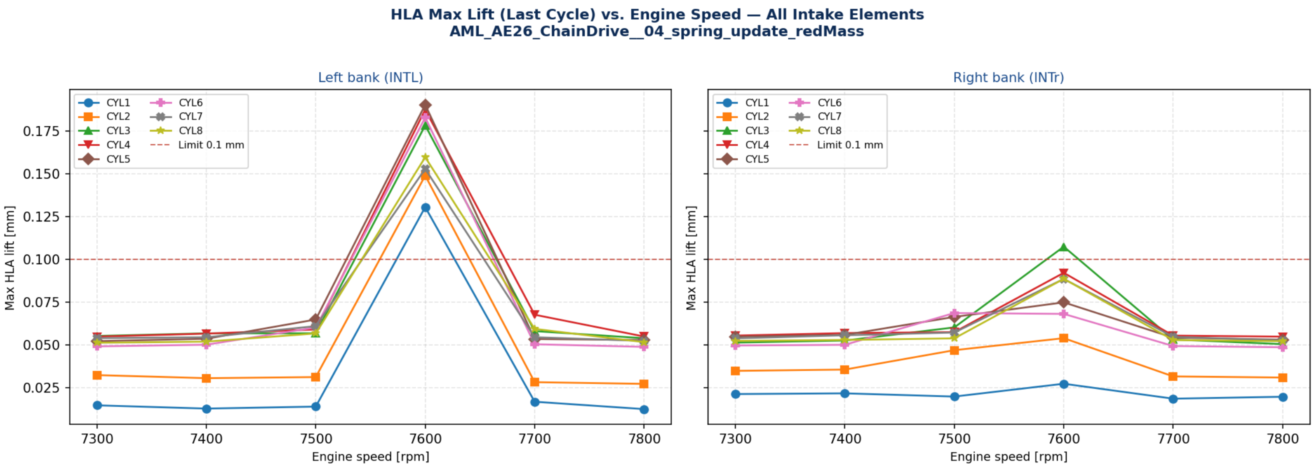

Fig. 5-3: Speed Sweep — Max Lift at Last Cycle *Max HLA lift in the last cam cycle vs. engine speed for all 16 intake elements. 7 speed points: 7000, 7300, 7400, 7500, 7600, 7700, 7800 rpm.*

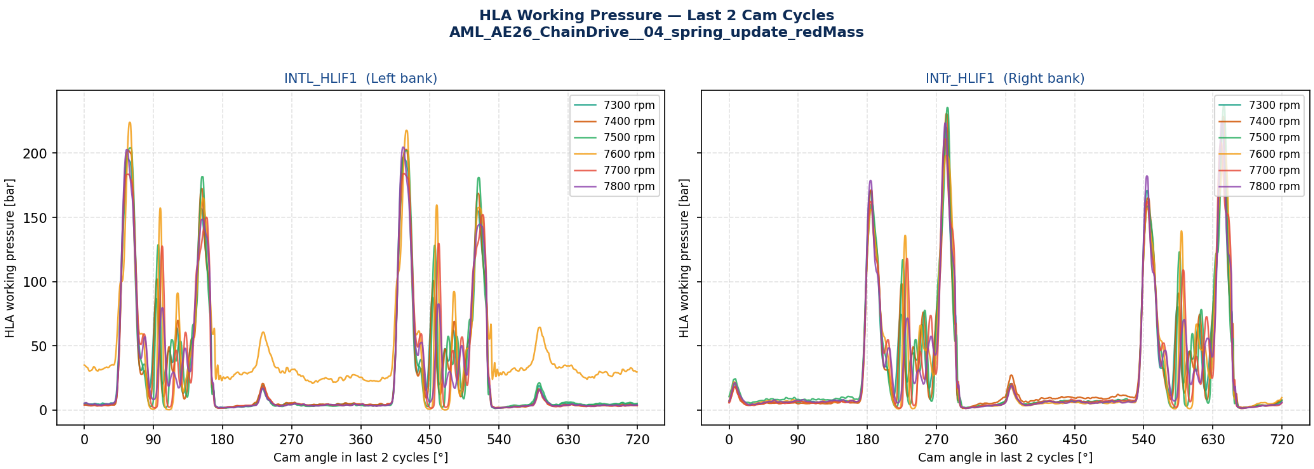

Fig. 5-4: HLA Working Pressure — Last 2 Cam Cycles *HLA working pressure over last 2 cycles. Elevated base-circle pressure corroborates pump-up detected via the lift signal.*

5.3 Pump-Up Summary — All Intake Elements

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?	Max wk. press. [bar]
INTL_HLIF1	7300	0.0145	-0.0844	no	196.4
INTL_HLIF2	7300	0.0322	-0.0711	no	199.2
INTL_HLIF3	7300	0.0550	-0.0473	no	195.2
INTL_HLIF4	7300	0.0547	-0.0353	no	191.7
INTL_HLIF5	7300	0.0520	-0.0497	no	195.4
INTL_HLIF6	7300	0.0490	-0.0504	no	195.8
INTL_HLIF7	7300	0.0538	-0.0465	no	197.2
INTL_HLIF8	7300	0.0511	-0.0494	no	195.9
INTr_HLIF1	7300	0.0211	-0.0897	no	219.7
INTr_HLIF2	7300	0.0347	-0.0733	no	213.5
INTr_HLIF3	7300	0.0512	-0.0630	no	225.2
INTr_HLIF4	7300	0.0553	-0.0549	no	225.8
INTr_HLIF5	7300	0.0547	-0.0507	no	213.5
INTr_HLIF6	7300	0.0495	-0.0616	no	213.0
INTr_HLIF7	7300	0.0537	-0.0566	no	207.3
INTr_HLIF8	7300	0.0522	-0.0535	no	212.3
INTL_HLIF1	7400	0.0126	-0.0847	no	202.8
INTL_HLIF2	7400	0.0304	-0.0722	no	201.0
INTL_HLIF3	7400	0.0566	-0.0427	no	199.1
INTL_HLIF4	7400	0.0564	-0.0429	no	196.6
INTL_HLIF5	7400	0.0534	-0.0491	no	199.1
INTL_HLIF6	7400	0.0500	-0.0549	no	200.5
INTL_HLIF7	7400	0.0543	-0.0541	no	206.3
INTL_HLIF8	7400	0.0518	-0.0536	no	206.0
INTr_HLIF1	7400	0.0215	-0.0843	no	228.6
INTr_HLIF2	7400	0.0354	-0.0741	no	221.7
INTr_HLIF3	7400	0.0524	-0.0604	no	232.7
INTr_HLIF4	7400	0.0567	-0.0489	no	227.8
INTr_HLIF5	7400	0.0557	-0.0603	no	223.5
INTr_HLIF6	7400	0.0499	-0.0604	no	222.2
INTr_HLIF7	7400	0.0555	-0.0456	no	204.6
INTr_HLIF8	7400	0.0527	-0.0536	no	207.8
INTL_HLIF1	7500	0.0138	-0.0909	no	201.9
INTL_HLIF2	7500	0.0310	-0.0749	no	201.8
INTL_HLIF3	7500	0.0566	-0.0483	no	198.9
INTL_HLIF4	7500	0.0589	-0.0401	no	207.6

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?	Max wk. press. [bar]
INTL_HLIF5	7500	0.0647	-0.0469	no	201.4
INTL_HLIF6	7500	0.0600	-0.0529	no	200.4
INTL_HLIF7	7500	0.0608	-0.0473	no	202.0
INTL_HLIF8	7500	0.0565	-0.0450	no	202.4
INTr_HLIF1	7500	0.0196	-0.0953	no	237.0
INTr_HLIF2	7500	0.0467	-0.0754	no	212.6
INTr_HLIF3	7500	0.0601	-0.0495	no	222.7
INTr_HLIF4	7500	0.0573	-0.0562	no	221.7
INTr_HLIF5	7500	0.0662	-0.0607	no	230.0
INTr_HLIF6	7500	0.0685	-0.0629	no	229.3
INTr_HLIF7	7500	0.0571	-0.0577	no	211.7
INTr_HLIF8	7500	0.0537	-0.0511	no	215.2
INTL_HLIF1	7600	0.1304	-0.0329	no	217.7
INTL_HLIF2	7600	0.1488	-0.0007	no	224.3
INTL_HLIF3	7600	0.1782	0.0035	no	222.4
INTL_HLIF4	7600	0.1874	0.0046	no	222.6
INTL_HLIF5	7600	0.1902	0.0111	no	220.7
INTL_HLIF6	7600	0.1828	0.0070	no	220.7
INTL_HLIF7	7600	0.1527	-0.0081	no	220.9
INTL_HLIF8	7600	0.1596	-0.0101	no	217.5
INTr_HLIF1	7600	0.0271	-0.0890	no	201.6
INTr_HLIF2	7600	0.0538	-0.0690	no	200.9
INTr_HLIF3	7600	0.1073	-0.0435	no	199.2
INTr_HLIF4	7600	0.0920	-0.0436	no	183.8
INTr_HLIF5	7600	0.0747	-0.0539	no	207.8
INTr_HLIF6	7600	0.0680	-0.0586	no	208.7
INTr_HLIF7	7600	0.0884	-0.0442	no	182.3
INTr_HLIF8	7600	0.0885	-0.0497	no	196.7
INTL_HLIF1	7700	0.0166	-0.0814	no	184.1
INTL_HLIF2	7700	0.0280	-0.0730	no	189.6
INTL_HLIF3	7700	0.0579	-0.0377	no	180.0
INTL_HLIF4	7700	0.0675	-0.0434	no	182.9
INTL_HLIF5	7700	0.0533	-0.0454	no	182.7
INTL_HLIF6	7700	0.0502	-0.0454	no	184.7
INTL_HLIF7	7700	0.0543	-0.0422	no	191.2
INTL_HLIF8	7700	0.0592	-0.0477	no	185.3
INTr_HLIF1	7700	0.0184	-0.0872	no	208.1

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?	Max wk. press. [bar]
INTr_HLIF2	7700	0.0314	-0.0802	no	210.9
INTr_HLIF3	7700	0.0530	-0.0609	no	215.8
INTr_HLIF4	7700	0.0553	-0.0527	no	210.3
INTr_HLIF5	7700	0.0541	-0.0561	no	215.3
INTr_HLIF6	7700	0.0492	-0.0602	no	216.7
INTr_HLIF7	7700	0.0546	-0.0527	no	195.1
INTr_HLIF8	7700	0.0527	-0.0528	no	216.5
INTL_HLIF1	7800	0.0123	-0.0886	no	204.6
INTL_HLIF2	7800	0.0271	-0.0761	no	209.3
INTL_HLIF3	7800	0.0537	-0.0465	no	201.3
INTL_HLIF4	7800	0.0547	-0.0360	no	195.3
INTL_HLIF5	7800	0.0527	-0.0524	no	203.6
INTL_HLIF6	7800	0.0487	-0.0474	no	197.2
INTL_HLIF7	7800	0.0522	-0.0398	no	190.4
INTL_HLIF8	7800	0.0514	-0.0428	no	196.8
INTr_HLIF1	7800	0.0195	-0.0933	no	224.7
INTr_HLIF2	7800	0.0308	-0.0815	no	222.5
INTr_HLIF3	7800	0.0503	-0.0507	no	211.1
INTr_HLIF4	7800	0.0547	-0.0535	no	205.4
INTr_HLIF5	7800	0.0529	-0.0545	no	214.8
INTr_HLIF6	7800	0.0485	-0.0584	no	214.5
INTr_HLIF7	7800	0.0525	-0.0447	no	197.9
INTr_HLIF8	7800	0.0520	-0.0542	no	213.0

★ 0 of 96 element-speed combinations exceed the 0.1 mm pump-up threshold (last cam cycle).

5.4 Interpretation and Engineering Assessment

The EXCITE TD dynamic simulation of the **AML_AE26_ChainDrive__04_spring_update_redMass** model covers seven speed points (7000, 7300, 7400, 7500, 7600, 7700, 7800 rpm) over 15 cam cycles (5400°) each — an extended run providing better convergence than the predecessor model. Results are evaluated at the last complete cam cycle.

No pump-up detected in any intake HLA element across the full speed range (7000–7800 rpm). The reduced-mass variant maintains HLA plunger lift below the 0.1 mm threshold at all conditions, confirming that the follower mass reduction is beneficial for HLA stability at high engine speeds.

Analysis performed on AVL EXCITE TD R2024.1 results — AML_AE26_ChainDrive__04_spring_update_redMass.etd. Results auto-generated by analyze_hlif_redmass.py — 2026-06-19.